LoginPage.java

```
package School_Management_System;

import java.awt.*;
import java.awt.event.*;
import javax.swing.*;
import java.sql.*;
public class LoginPage extends JFrame implements ActionListener {

    JLayeredPane lp1;
    JLabel l1, l2, l3, l4, bg1;
    JTextField tf1;
    JPasswordField pf1;
    Choice ch1;
    JButton bt1, bt2;
    Image il;

    LoginPage() {
        setTitle("Login - School Management System");
        setSize(750, 460);
        setResizable(false);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLocationRelativeTo(null);

        ImageIcon img = new
ImageIcon(getClass().getResource("/School_Management_System/Icons/background.jpg"));
        Image il=img.getImage().getScaledInstance(750, 460, Image.SCALE_SMOOTH);

        //background image work
        bg1 = new JLabel();
        bg1.setIcon(new ImageIcon(il));
        bg1.setBounds(0, 0, 750, 460);


        lp1 = new JLayeredPane();
        lp1.setBounds(0, 0, 750, 460);
        getContentPane().setLayout(null);
        getContentPane().add(lp1);

        JPanel l0 = new JPanel(null);
        l0.setOpaque(false);
        l0.setBounds(0, 0, 750, 460);

        //main header stating login
```

```
l1 = new JLabel("Login School Account");
l1.setBounds(200, 40, 400, 40);
l1.setFont(new Font("Arial", Font.BOLD, 28));
l1.setForeground(Color.WHITE);
l0.add(l1);

//account text in orange
l2 = new JLabel("Account:");
l2.setBounds(150, 120, 150, 30);
l2.setFont(new Font("Arial", Font.BOLD, 20));
l2.setForeground(new Color(245, 135, 66));
l0.add(l2);

//dropdown of account type
ch1 = new Choice();
ch1.add("Select Account Type");
ch1.add("Admin");
ch1.add("Teacher");
ch1.add("Student");
ch1.setFont(new Font("Arial", Font.BOLD, 18));
ch1.setBounds(330, 120, 200, 30);
l0.add(ch1);

//text in orange of username
l3 = new JLabel("Username:");
l3.setBounds(150, 180, 150, 30);
l3.setFont(new Font("Arial", Font.BOLD, 20));
l3.setForeground(new Color(245, 135, 66));
l0.add(l3);

//textbox for username input
tf1 = new JTextField();
tf1.setFont(new Font("Arial", Font.PLAIN, 20));
tf1.setBounds(330, 180, 200, 30);
l0.add(tf1);

//password text in orange
l4 = new JLabel("Password:");
l4.setBounds(150, 240, 150, 30);
l4.setFont(new Font("Arial", Font.BOLD, 20));
l4.setForeground(new Color(245, 135, 66));
l0.add(l4);

//password box for inputing password values
```

```
        pf1 = new JPasswordField();
        pf1.setFont(new Font("Arial", Font.PLAIN, 20));
        pf1.setBounds(330, 240, 200, 30);
        l0.add(pf1);

        //buttons of login/exit
        bt1 = new JButton("Login");
        bt2 = new JButton("Exit");
        l0.add(bt1);
        l0.add(bt2);

        //design of login button
        bt1.setFont(new Font("Arial", Font.BOLD, 18));
        bt2.setFont(new Font("Arial", Font.BOLD, 18));
        bt1.setBounds(150, 320, 150, 40);
        bt2.setBounds(380, 320, 150, 40);
        bt1.setBackground(new Color(53, 4, 117));
        bt1.setForeground(Color.WHITE);
        bt2.setBackground(new Color(191, 247, 161));
        bt2.setForeground(Color.BLACK);

        //acctions of login&exit
        bt1.addActionListener(this);
        bt2.addActionListener(this);

        lp1.add(bg1, Integer.valueOf(0));
        lp1.add(l0, Integer.valueOf(1));

        setVisible(true);
    }

    @Override
  public void actionPerformed(ActionEvent ae) {
    if (ae.getSource() == bt1) {
        try {
            ConnectionClass obj = new ConnectionClass();
            String account = ch1.getSelectedItem();
            String username = tf1.getText();
            String password = pf1.getText();
            if (account.equals("Select Account Type")) {
                JOptionPane.showMessageDialog(null, "You have not Selected an Account Type");
                tf1.setText("");
                pf1.setText("");
                new LoginPage();
```

```
        }
        else if (account.equals("Admin")) {
                String q = "select * from admin where username='" + username + "' and
password='" + password + "'";
                ResultSet rest = obj.stm.executeQuery(q);
                if (rest.next())
                {
                    new AdminHomePage(account, rest.getString("username")).setVisible(true);
                    System.out.println("Admin Login Successful");
                }else
                {
                    JOptionPane.showMessageDialog(null, "You have entered Wrong Username
and Password!");
                    tf1.setText("");
                    pf1.setText("");
                    new LoginPage();
                }
            }
            else if (account.equals("Teacher")) {
                String q = "select * from teacher where username='" + username + "' ";
                        //+ "and password='" + password + "'";
                ResultSet rest = obj.stm.executeQuery(q);
                if (rest.next())
                {
                    String PSS=rest.getString("password");
                    boolean passmatch=PasswordUtils.verifyPassword(PSS, password);
                    if (passmatch)
                    {
                        new TeacherHomePage(rest.getString("username"),account);
                        System.out.println("Teacher Login Successful");
                    }
                    else
                    {
                        JOptionPane.showMessageDialog(null, "You have entered Wrong
Password!");
                        tf1.setText("");
                        pf1.setText("");
                        new LoginPage();
                    }
                }else
                {
                    JOptionPane.showMessageDialog(null, "You have entered Wrong Username
and Password!");
                    tf1.setText("");
```

```
                    pf1.setText("");
                    new LoginPage();
                }
            } else if (account.equals("Student")) {
                String q = "select * from student where username='" + username + "'";
//                        + "and password='" + password + "'";
                ResultSet rest = obj.stm.executeQuery(q);
                if (rest.next())
                {
                    String PSS=rest.getString("password");
                    boolean passmatch=PasswordUtils.verifyPassword(PSS, password);
                    if (passmatch)
                    {
                        new StudentHomePage(account, rest.getString("username")).setVisible(true);
                        System.out.println("Teacher Login Successful");
                    }
                    else
                    {
                        JOptionPane.showMessageDialog(null, "You have entered Wrong
Password!");
                        tf1.setText("");
                        pf1.setText("");
                        new LoginPage();
                    }
                }else
                {
                    JOptionPane.showMessageDialog(null, "You have entered Wrong Username
and Password!");
                    tf1.setText("");
                    pf1.setText("");
                    new LoginPage();
                }
            }
            this.setVisible(false);

        } catch (Exception ex) {
            ex.printStackTrace();
        }
    }

    if (ae.getSource() == bt2) {
        System.exit(0);
    }
}
```

```
    public static void main(String[] args)
    {
        new LoginPage();
    }
}
```

AdminHomePage.java

```
package School_Management_System;

import java.awt.event.*;
import java.awt.*;
import javax.swing.*;

public class AdminHomePage extends JFrame implements ActionListener {
   Font f, f1;
   public String id, account2, pub_username;

   AdminHomePage(String account, String username)
   {
      account2 = account;
      pub_username=username;

      this.setTitle("School Management System ADMIN HOME PAGE");
      setLocation(0, 0);
      setSize(2000, 1050);
      setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
      setResizable(false);

      f = new Font("Arial", Font.BOLD, 20);
      f1 = new Font("Arial", Font.BOLD, 18);

      // Load image
      ImageIcon img = new
ImageIcon(getClass().getResource("/School_Management_System/Icons/adminhome.jpg"));
      Image scaledImg = img.getImage().getScaledInstance(2000, 1050,
Image.SCALE_SMOOTH);

      // Set background label
      JLabel bg1 = new JLabel(new ImageIcon(scaledImg));
      bg1.setBounds(0, 0, 2000, 1050);

      // Add background first
      getContentPane().add(bg1);

      // Optional: add layered panel on top
      JLayeredPane layeredPane = new JLayeredPane();
      layeredPane.setBounds(0, 0, 2000, 1200);
      layeredPane.setOpaque(false);
      getContentPane().add(layeredPane);
```

```
JMenuBar m1 = new JMenuBar();
m1.setBackground(Color.black.darker());

//menu buttons extender
JMenu menu1 = new JMenu("Teacher Profile");
JMenu menu2 = new JMenu("Student Profile");
JMenu menu3 = new JMenu("Class Details");
JMenu menu4 = new JMenu("Subject Details");
JMenu menu5 = new JMenu("Marks Details");
JMenu menu6 = new JMenu("Fee Details");
JMenu menu7 = new JMenu("Result");
JMenu menu8 = new JMenu("Logout");

// teacher menu items
JMenuItem ment1 = new JMenuItem("Add Teacher Details");
JMenuItem ment2 = new JMenuItem("Update Teacher Details");
JMenuItem ment3 = new JMenuItem("View Teacher Details");

// student menu items
JMenuItem ment4 = new JMenuItem("Add Student Details");
JMenuItem ment5 = new JMenuItem("Update Student Details");
JMenuItem ment6 = new JMenuItem("View Student Details");

// class menu items
JMenuItem ment7 = new JMenuItem("Add Class Details");
JMenuItem ment8 = new JMenuItem("Update Class Details");


// subject menu items
JMenuItem ment9 = new JMenuItem("Add Subject Details");

// marks mark items
JMenuItem ment10 = new JMenuItem("Add Marks");
JMenuItem ment11 = new JMenuItem("View Marks");

// fee menu items
JMenuItem ment12 = new JMenuItem("Add Fee Details");
JMenuItem ment13 = new JMenuItem("Add Fee Structure");
JMenuItem ment14 = new JMenuItem("View Fee Details");

// result menu items
JMenuItem ment15 = new JMenuItem("Show Results");

// logout menu items
```

```
JMenuItem ment16 = new JMenuItem("Exit");

// add items to menu1 Teacher Profile
menu1.add(ment1);
menu1.add(ment2);
menu1.add(ment3);

// add items to menu2 Student Profile
menu2.add(ment4);
menu2.add(ment5);
menu2.add(ment6);

// add items to menu3 Classes
menu3.add(ment7);
menu3.add(ment8);

// add items to menu4 Subject Details
menu4.add(ment9);

// add items to menu5 Marks
menu5.add(ment10);
menu5.add(ment11);

// add items to menu6 Fee Details
menu6.add(ment12);
menu6.add(ment13);
menu6.add(ment14);

// add items to menu7 Results
menu7.add(ment15);

// add items to menu8 Logout
menu8.add(ment16);

// add menus to menubar
m1.add(menu1);
m1.add(menu2);
m1.add(menu3);
m1.add(menu4);
m1.add(menu5);
m1.add(menu6);
m1.add(menu7);
m1.add(menu8);
```

```
//menu
menu1.setFont(f);
menu2.setFont(f);
menu3.setFont(f);
menu4.setFont(f);
menu5.setFont(f);
menu6.setFont(f);
menu7.setFont(f);
menu8.setFont(f);

ment1.setFont(f1);
ment2.setFont(f1);
ment3.setFont(f1);
ment4.setFont(f1);
ment5.setFont(f1);
ment6.setFont(f1);
ment7.setFont(f1);
ment8.setFont(f1);
ment9.setFont(f1);
ment10.setFont(f1);
ment11.setFont(f1);
ment12.setFont(f1);
ment13.setFont(f1);
ment14.setFont(f1);
ment15.setFont(f1);
ment16.setFont(f1);

menu1.setForeground(Color.GRAY);
menu2.setForeground(Color.GRAY);
menu3.setForeground(Color.GRAY);
menu4.setForeground(Color.GRAY);
menu5.setForeground(Color.GRAY);
menu6.setForeground(Color.GRAY);
menu7.setForeground(Color.GRAY);
menu8.setForeground(Color.RED);

ment1.setForeground(Color.ORANGE);
ment2.setForeground(Color.ORANGE);
ment3.setForeground(Color.ORANGE);
ment4.setForeground(Color.ORANGE);
ment5.setForeground(Color.ORANGE);
ment6.setForeground(Color.ORANGE);
ment7.setForeground(Color.ORANGE);
ment8.setForeground(Color.ORANGE);
```

```
ment9.setForeground(Color.ORANGE);
ment10.setForeground(Color.ORANGE);
ment11.setForeground(Color.ORANGE);
ment12.setForeground(Color.ORANGE);
ment13.setForeground(Color.ORANGE);
ment14.setForeground(Color.ORANGE);
ment15.setForeground(Color.ORANGE);
ment16.setForeground(Color.RED);

ment1.setBackground(Color.BLACK);
ment2.setBackground(Color.BLACK);
ment3.setBackground(Color.BLACK);
ment4.setBackground(Color.BLACK);
ment5.setBackground(Color.BLACK);
ment6.setBackground(Color.BLACK);
ment7.setBackground(Color.BLACK);
ment8.setBackground(Color.BLACK);
ment9.setBackground(Color.BLACK);
ment10.setBackground(Color.BLACK);
ment11.setBackground(Color.BLACK);
ment12.setBackground(Color.BLACK);
ment13.setBackground(Color.BLACK);
ment14.setBackground(Color.BLACK);
ment15.setBackground(Color.BLACK);
ment16.setBackground(Color.BLACK);

ment1.addActionListener(this);
ment2.addActionListener(this);
ment3.addActionListener(this);
ment4.addActionListener(this);
ment5.addActionListener(this);
ment6.addActionListener(this);
ment7.addActionListener(this);
ment8.addActionListener(this);
ment9.addActionListener(this);
ment10.addActionListener(this);
ment11.addActionListener(this);
ment12.addActionListener(this);
ment13.addActionListener(this);
ment14.addActionListener(this);
ment15.addActionListener(this);
ment16.addActionListener(this);
```

```java
        setJMenuBar(m1);
    }

    @Override
    public void actionPerformed(ActionEvent ae)
    {
        String comnd=ae.getActionCommand();
        if (comnd.equals("Add Teacher Details"))
        {
//            System.out.println("Add Teacher Details class open");
            new AddTeacherDetails().setVisible(true);
        }
        else if (comnd.equals("Update Teacher Details"))
        {
//            System.out.println("Update Teacher Details class open");
            new UpdateTeacherDetails(account2, pub_username).setVisible(true);
        }
        else if (comnd.equals("View Teacher Details"))
        {
//            System.out.println("View Teacher Details class open");
            new ViewTeacherDetails(account2, pub_username).setVisible(true);
        }
        else if (comnd.equals("Add Student Details"))
        {
//            System.out.println("Add Student Details class open");
            new AddStudentDetails().setVisible(true);
        }
        else if (comnd.equals("Update Student Details"))
        {
//            System.out.println("Update Student Details class open");
            new UpdateStudentDetails(account2, pub_username).setVisible(true);
        }
        else if (comnd.equals("View Student Details"))
        {
//            System.out.println("View Student Details class open");
            new ViewStudentDetails(account2, pub_username).setVisible(true);
        }
        else if (comnd.equals("Add Class Details"))
        {
//            System.out.println("Add Class Details class open");
            new AddNewClass().setVisible(true);
        }
        else if (comnd.equals("Update Class Details"))
        {
```

```
//          System.out.println("Update Class Details class open");
          new UpdateClassDetails().setVisible(true);
        }
        else if (comnd.equals("Add Subject Details"))
        {
//          System.out.println("Add Subject Details class open");
          new AddSubjectDetails().setVisible(true);
        }
        else if (comnd.equals("Add Marks"))
        {
//          System.out.println("Add Mark Subject class open");
          new AddMarksDetails().setVisible(true);
        }
        else if (comnd.equals("View Marks"))
        {
//          System.out.println("Add Mark Subject class open");
          new ViewMarksDetails(pub_username, account2).setVisible(true);
        }
        else if (comnd.equals("Add Fee Details"))
        {
//          System.out.println("Add Fee Details class open");
          new AddFeeDetails().setVisible(true);
        }
        else if (comnd.equals("Add Fee Structure"))
        {
          System.out.println("Add Fee Structure class open");
          new AddFeeStructure().setVisible(true);
        }
        else if (comnd.equals("View Fee Details"))
        {
          System.out.println("View Fee Details class open");
          new ViewFeeDetails(pub_username, account2).setVisible(true);
        }
        else if (comnd.equals("Show Results"))
        {
          System.out.println("Show Results class open");
          new ShowResult(pub_username, account2).setVisible(true);
        }
        else if (comnd.equals("Exit"))
        {
          System.out.println("You are Logged Out");
          this.setVisible(false);
          new LoginPage();
        }
```

```
    }

//    public static void main (String[] args) {
//        new AdminHomePage("account", "username").setVisible(true);
//    }
}
```

TeacherHomePage.java

```
package School_Management_System;

import java.awt.event.*;
import java.awt.*;
import javax.swing.*;

public class TeacherHomePage extends JFrame implements ActionListener {
    JLabel l5;
    Font f, f1;
    public String id, account2, pub_username, classe;

    TeacherHomePage(String username, String account) {
        account2 = account;
        pub_username=username;
        this.setTitle("School Management System TEACHER HOME PAGE");
        setLocation(0, 0);
        setSize(2000, 1200);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setResizable(false);

        f = new Font("Arial", Font.BOLD, 20);
        f1 = new Font("Arial", Font.BOLD, 18);

        // Load background
        ImageIcon img = new
ImageIcon(getClass().getResource("/School_Management_System/Icons/teacherhome.jpg"));
        Image scaledImg = img.getImage().getScaledInstance(2000, 1050,
Image.SCALE_SMOOTH);
        JLabel bg1 = new JLabel(new ImageIcon(scaledImg));
        bg1.setBounds(0, 0, 2000, 1050);

        JLayeredPane layeredPane = new JLayeredPane();
        layeredPane.setBounds(0, 0, 2000, 1200);
        layeredPane.add(bg1, Integer.valueOf(0));

        getContentPane().add(layeredPane);

        // Menu bar setup
        JMenuBar m1 = new JMenuBar();
        setJMenuBar(m1);
        m1.setBackground(Color.black.darker());

        setVisible(true);
```

```
//menu buttons extender
JMenu menu1 = new JMenu("Teacher Profile");
JMenu menu2 = new JMenu("Student Profile");
JMenu menu3 = new JMenu("Marks Details");
JMenu menu4 = new JMenu("Result");
JMenu menu5 = new JMenu("Logout");

// teacher menu items
JMenuItem ment1 = new JMenuItem("Update Teacher Details");
JMenuItem ment2 = new JMenuItem("View Teacher Details");

// student menu items
JMenuItem ment3 = new JMenuItem("Update Student Details");
JMenuItem ment4 = new JMenuItem("View Student Details");

// marks mark items
JMenuItem ment5 = new JMenuItem("Add Mark Subject");
JMenuItem ment8 = new JMenuItem("View Mark Details");

// result menu items
JMenuItem ment6 = new JMenuItem("Show Results");

// logout menu items
JMenuItem ment7 = new JMenuItem("Exit");

//add buttons to menus
menu1.add(ment1);
menu1.add(ment2);
menu2.add(ment3);
menu2.add(ment4);
menu3.add(ment5);
menu4.add(ment6);
menu5.add(ment7);

// add menus to menubar
m1.add(menu1);
m1.add(menu2);
m1.add(menu3);
m1.add(menu4);
m1.add(menu5);

 //menu items customization
menu1.setFont(f);
```

```
menu2.setFont(f);
menu3.setFont(f);
menu4.setFont(f);
menu5.setFont(f);

ment1.setFont(f1);
ment2.setFont(f1);
ment3.setFont(f1);
ment4.setFont(f1);
ment5.setFont(f1);
ment6.setFont(f1);
ment7.setFont(f1);

menu1.setForeground(Color.GRAY);
menu2.setForeground(Color.GRAY);
menu3.setForeground(Color.GRAY);
menu4.setForeground(Color.GRAY);
menu5.setForeground(Color.RED);

ment1.setForeground(Color.ORANGE);
ment2.setForeground(Color.ORANGE);
ment3.setForeground(Color.ORANGE);
ment4.setForeground(Color.ORANGE);
ment5.setForeground(Color.ORANGE);
ment6.setForeground(Color.ORANGE);
ment7.setForeground(Color.RED);

ment1.setBackground(Color.BLACK);
ment2.setBackground(Color.BLACK);
ment3.setBackground(Color.BLACK);
ment4.setBackground(Color.BLACK);
ment5.setBackground(Color.BLACK);
ment6.setBackground(Color.BLACK);
ment7.setBackground(Color.BLACK);

ment1.addActionListener(this);
ment2.addActionListener(this);
ment3.addActionListener(this);
ment4.addActionListener(this);
ment5.addActionListener(this);
ment6.addActionListener(this);
ment7.addActionListener(this);

setJMenuBar(m1);
```

```
    }

    @Override
    public void actionPerformed(ActionEvent ae)
    {
        String comnd = ae.getActionCommand();
        if (comnd.equals("Update Teacher Details"))
        {
            System.out.println("Update Teacher Details class open");
            new UpdateTeacherDetails(account2, pub_username).setVisible(true);
        }
        else if (comnd.equals("View Teacher Details"))
        {
//            System.out.println("View Teacher Details class open");
            new ViewTeacherDetails(account2, pub_username).setVisible(true);
        }
        else if (comnd.equals("Update Student Details"))
        {
//            System.out.println("Update Student Details class open");
            new UpdateStudentDetails(account2, pub_username).setVisible(true);
        }
        else if (comnd.equals("View Student Details"))
        {
//            System.out.println("View Student Details class open");
            new ViewStudentDetails(account2, pub_username).setVisible(true);
        }
        else if (comnd.equals("Update Class Details"))
        {
//            System.out.println("Update Class Details class open");
            new UpdateClassDetails().setVisible(true);
        }
        else if (comnd.equals("Add Mark Subject"))
        {
//            System.out.println("Add Mark Subject class open");
            new AddMarksDetails().setVisible(true);
        }
        else if (comnd.equals("Show Results"))
        {
//            System.out.println("Show Results class open");
            new ShowResult(pub_username, account2).setVisible(true);
        }
        else if (comnd.equals("Exit"))
        {
```

```
            System.out.println("You are Logged Out");
            this.setVisible(false);
            new LoginPage();
        }
    }
//     public static void main (String[] args) {
//        new TeacherHomePage("account2", "pub_username").setVisible(true);
//    }
}
```

StudentHomePage.java

```
package School_Management_System;

import java.awt.event.*;
import java.awt.*;
import javax.swing.*;

public class StudentHomePage extends JFrame implements ActionListener {
    JLabel l5;
    Font f, f1;
    public String account2, pub_username, id;

    StudentHomePage(String username, String account)
    {

        account2 = account;
        pub_username = username;

        setTitle("School Management System STUDENT HOME PAGE");
        setLocation(0, 0);
        setSize(2000, 1200);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setResizable(false);

        f = new Font("Arial", Font.BOLD, 20);
        f1 = new Font("Arial", Font.BOLD, 18);

        // Load background
        ImageIcon img = new ImageIcon(getClass().getResource("Icons/studenthome.jpg"));
        Image scaledImg = img.getImage().getScaledInstance(2000, 1050,
Image.SCALE_SMOOTH);
        JLabel bg1 = new JLabel(new ImageIcon(scaledImg));
        bg1.setBounds(0, 0, 2000, 1050);

        JLayeredPane layeredPane = new JLayeredPane();
        layeredPane.setBounds(0, 0, 2000, 1200);
        layeredPane.add(bg1, Integer.valueOf(0));

        getContentPane().add(layeredPane);

        // Menu bar setup
        JMenuBar m1 = new JMenuBar();
        setJMenuBar(m1);
        m1.setBackground(Color.black.darker());
```

```
setVisible(true);

//menu buttons extender
JMenu menu1 = new JMenu("Teacher Profile");
JMenu menu2 = new JMenu("Student Profile");
JMenu menu3 = new JMenu("Marks Details");
JMenu menu4 = new JMenu("Result");
JMenu menu5 = new JMenu("Logout");

// student menu items
JMenuItem ment1 = new JMenuItem("View Teacher Details");

JMenuItem ment2 = new JMenuItem("Update Student Details");
JMenuItem ment3 = new JMenuItem("View Student Details");

JMenuItem ment4 = new JMenuItem("View Marks Details");

// result menu items
JMenuItem ment5 = new JMenuItem("Show Results");

// logout menu items
JMenuItem ment6 = new JMenuItem("Exit");

//add buttons to menus
menu1.add(ment1);
menu2.add(ment2);
menu2.add(ment3);
menu3.add(ment4);
menu4.add(ment5);
menu5.add(ment6);

// add menus to menubar
m1.add(menu1);
m1.add(menu2);
m1.add(menu3);
m1.add(menu4);
m1.add(menu5);

 //menu items customization
menu1.setFont(f);
menu2.setFont(f);
menu3.setFont(f);
menu4.setFont(f);
```

```
    menu5.setFont(f);

    ment1.setFont(f1);
    ment2.setFont(f1);
    ment3.setFont(f1);
    ment4.setFont(f1);
    ment5.setFont(f1);
    ment6.setFont(f1);

    menu1.setForeground(Color.GRAY);
    menu2.setForeground(Color.GRAY);
    menu3.setForeground(Color.GRAY);
    menu4.setForeground(Color.GRAY);
    menu5.setForeground(Color.RED);

    ment1.setForeground(Color.ORANGE);
    ment2.setForeground(Color.ORANGE);
    ment3.setForeground(Color.ORANGE);
    ment4.setForeground(Color.ORANGE);
    ment5.setForeground(Color.ORANGE);
    ment6.setForeground(Color.RED);

    ment1.setBackground(Color.BLACK);
    ment2.setBackground(Color.BLACK);
    ment3.setBackground(Color.BLACK);
    ment4.setBackground(Color.BLACK);
    ment5.setBackground(Color.BLACK);
    ment6.setBackground(Color.BLACK);

    ment1.addActionListener(this);
    ment2.addActionListener(this);
    ment3.addActionListener(this);
    ment4.addActionListener(this);
    ment5.addActionListener(this);
    ment6.addActionListener(this);

    setJMenuBar(m1);

}

@Override
public void actionPerformed(ActionEvent ae)
{
    String comnd = ae.getActionCommand();
```

```
        if (comnd.equals("Update Student Details"))
        {
            System.out.println("Update Student Details class open");
            //new UpdateStudentDetails().setVisible(true);
        }
        else if (comnd.equals("View Student Details"))
        {
//            System.out.println("View Student Details class open");
            new ViewStudentDetails(pub_username, account2).setVisible(true);
        }
        else if (comnd.equals("View Teacher Details"))
        {
//            System.out.println("View Teacher Details class open");
            new ViewTeacherDetails(account2, pub_username).setVisible(true);
        }
        else if (comnd.equals("View Marks Details"))
        {
            System.out.println("View Marks Details class open");
            //new ViewMarksDetails().setVisible(true);
        }
        else if (comnd.equals("Show Results"))
        {
            System.out.println("Show Results class open");
            new ShowResult("account2", "pub_username").setVisible(true);
        }
        else if (comnd.equals("Exit"))
        {
            System.out.println("You are Logged Out");
            this.setVisible(false);
            new LoginPage();
        }
    }
//    public static void main (String[] args)
//    {
//        new StudentHomePage("account2", "pub_username").setVisible(true);
//    }
}
```

AddTeacherDetails.java

```
package School_Management_System;

import java.awt.*;
import javax.swing.*;
import java.awt.event.*;
import java.security.NoSuchAlgorithmException;
import java.util.Random;

public class AddTeacherDetails extends JFrame implements ActionListener
{

   JLabel l1, l2, l3, l4, l5, l6, l7, l8, l9, l10, l11, l12, l13;
   JButton bt1, bt2;
   JTextField tf1, tf2, tf3, tf4, tf5, tf6, tf7, tf8, tf9, tf10, tf11;
   JPasswordField pf1;

   AddTeacherDetails()
   {
      setTitle("Add New Teacher Details");
      getContentPane().setBackground(Color.WHITE);
      setLayout(null);
      setSize(840, 600);
      setResizable(false);
      setLocation(550,200);
      setVisible(true);


      // Load background
      ImageIcon img = new ImageIcon(getClass().getResource("Icons/teacheradd.jpg"));
      Image scaledImg = img.getImage().getScaledInstance(840, 600,
Image.SCALE_SMOOTH);
      JLabel bg1 = new JLabel(new ImageIcon(scaledImg));
      bg1.setBounds(0, 0, 840, 600);
      bg1.setLayout(null);
      add(bg1);

      l1= new JLabel("Add Teacher Details for Signup");
      l1.setBounds(230, 50, 500, 50);
      l1.setFont(new Font("Ariel", Font.BOLD, 30));
      l1.setForeground(new Color(8, 161, 92));
      bg1.add(l1);

      l2= new JLabel("Name");
```

```
l2.setBounds(50, 150, 150, 30);
l2.setFont(new Font("Ariel", Font.BOLD, 20));
l2.setForeground(Color.WHITE);
bg1.add(l2);

tf1= new JTextField();
tf1.setBounds(200, 150, 150, 30);
bg1.add(tf1);

l3= new JLabel("Username");
l3.setBounds(450, 150, 150, 30);
l3.setFont(new Font("Ariel", Font.BOLD, 20));
l3.setForeground(Color.WHITE);
bg1.add(l3);

tf2= new JTextField();
tf2.setBounds(600, 150, 150, 30);
bg1.add(tf2);

l4= new JLabel("Password");
l4.setBounds(50, 200, 150, 30);
l4.setFont(new Font("Ariel", Font.BOLD, 20));
l4.setForeground(Color.WHITE);
bg1.add(l4);

pf1= new JPasswordField();
pf1.setBounds(200, 200, 150, 30);
bg1.add(pf1);

l5= new JLabel("E-Mail");
l5.setBounds(450, 200, 150, 30);
l5.setFont(new Font("Ariel", Font.BOLD, 20));
l5.setForeground(Color.WHITE);
bg1.add(l5);

tf3= new JTextField();
tf3.setBounds(600, 200, 150, 30);
bg1.add(tf3);

l6= new JLabel("Father's name");
l6.setBounds(50, 250, 150, 30);
l6.setFont(new Font("Ariel", Font.BOLD, 20));
l6.setForeground(Color.WHITE);
bg1.add(l6);
```

```
tf4= new JTextField();
tf4.setBounds(200, 250, 150, 30);
bg1.add(tf4);

l7= new JLabel("Phone");
l7.setBounds(450, 250, 150, 30);
l7.setFont(new Font("Ariel", Font.BOLD, 20));
l7.setForeground(Color.WHITE);
bg1.add(l7);

tf5= new JTextField();
tf5.setBounds(600, 250, 150, 30);
bg1.add(tf5);

l8= new JLabel("Blood Group");
l8.setBounds(50, 300, 150, 30);
l8.setFont(new Font("Ariel", Font.BOLD, 20));
l8.setForeground(Color.WHITE);
bg1.add(l8);

tf6= new JTextField();
tf6.setBounds(200, 300, 150, 30);
bg1.add(tf6);

l9= new JLabel("Gender");
l9.setBounds(450, 300, 150, 30);
l9.setFont(new Font("Ariel", Font.BOLD, 20));
l9.setForeground(Color.WHITE);
bg1.add(l9);

tf7= new JTextField();
tf7.setBounds(600, 300, 150, 30);
bg1.add(tf7);

l10= new JLabel("City");
l10.setBounds(50, 350, 150, 30);
l10.setFont(new Font("Ariel", Font.BOLD, 20));
l10.setForeground(Color.WHITE);
bg1.add(l10);

tf8= new JTextField();
tf8.setBounds(200, 350, 150, 30);
bg1.add(tf8);
```

```
l11= new JLabel("Age");
l11.setBounds(450, 350, 150, 30);
l11.setFont(new Font("Ariel", Font.BOLD, 20));
l11.setForeground(Color.WHITE);
bg1.add(l11);

tf9= new JTextField();
tf9.setBounds(600, 350, 150, 30);
bg1.add(tf9);

l12= new JLabel("Teaching Experience");
l12.setBounds(2, 400, 200, 30);
l12.setFont(new Font("Ariel", Font.BOLD, 20));
l12.setForeground(Color.WHITE);
bg1.add(l12);

tf10= new JTextField();
tf10.setBounds(200, 400, 150, 30);
bg1.add(tf10);

l13= new JLabel("DOB");
l13.setBounds(450, 400, 150, 30);
l13.setFont(new Font("Ariel", Font.BOLD, 20));
l13.setForeground(Color.WHITE);
bg1.add(l13);

tf11= new JTextField();
tf11.setBounds(600, 400, 150, 30);
bg1.add(tf11);

//buttons
bt1= new JButton("Sumbit");
bt1.setBackground(Color.BLACK);
bt1.setForeground(Color.WHITE);
bt1.setBounds(250, 500, 150, 30);

bt2= new JButton("Cancel");
bt2.setBackground(new Color(8, 161, 92));
bt2.setForeground(Color.BLACK);
bt2.setBounds(500, 500, 150, 30);

bg1.add(bt1);
bg1.add(bt2);
```

```
    bt1.addActionListener(this);
    bt2.addActionListener(this);

    //Force UI refresh and visibility

    revalidate();
    repaint();
    setVisible(true);

}

@Override
public void actionPerformed(ActionEvent ae)
{
    if(ae.getSource()==bt1)
    {
        String name=tf1.getText();
        String username=tf2.getText();
        String password=pf1.getText();
        // Hash the password with a salt
        String hashedPassword=null;
            try {
                hashedPassword = PasswordUtils.hashPassword(password);
            }
            catch (NoSuchAlgorithmException ex) { }
                //Logger.getLogger(UserScreen.class.getName()).log(Level.SEVERE, null, ex);
        String email=tf3.getText();
        String father_name=tf4.getText();
        String phone=tf5.getText();
        String blood_group=tf6.getText();
        String gender=tf7.getText();
        String city=tf8.getText();
        String age=tf9.getText();
        String teaching_exp=tf10.getText();
        String dob=tf11.getText();
        Random r= new Random();
        String tec_id=""+Math.abs(r.nextInt() % 100000);

        try
        {
            ConnectionClass obj= new ConnectionClass();
```

```
            String q= "insert into teacher
values('"+tec_id+"','"+name+"','"+username+"','"+hashedPassword+"','"+email+"','"+father_name
+"','"+phone+"','"+blood_group+"','"+gender+"','"+city+"','"+age+"','"+teaching_exp+"','"+dob+"')";
            obj.stm.executeUpdate(q);
            JOptionPane.showMessageDialog(null,"Details Successfully Inserted");
            setVisible(false);
         }
         catch(Exception ex)
         {
            ex.printStackTrace();
         }
      }
      if (ae.getSource()==bt2)
      {
         setVisible(false);
      }
   }
//    public static void main(String[] args)
//    {
//       new AddTeacherDetails().setVisible(true);
//    }
}
```

UpdateTeacherDetails.java

```
package School_Management_System;

import java.awt.*;
import javax.swing.*;
import java.awt.event.*;
import java.sql.*;

public class UpdateTeacherDetails extends JFrame implements ActionListener
{
   JLabel l1, l2, l3, l4, l5, l6, l7, l8, l9, l10, l11, l12;
   JPanel p1, p2, p3;
   JTextField tf1, tf2, tf3, tf4, tf5, tf6, tf7, tf8, tf9, tf10;
   Choice ch1;
   JButton bt1, bt2;
   public String pubu, account2;

   UpdateTeacherDetails(String account, String username)
   {
      super("Update Teacher Details");
      setSize(760, 720);
      setLocation(550,200);
      setVisible(true);
      pubu=username;
      account2=account;

      Font f = new Font("Arial", Font.BOLD, 28);
      Font f1 = new Font("Arial", Font.PLAIN, 18);

      ch1=new Choice();
      ch1.setFont(f1);
      ch1.add("Select Username");

      try
      {
         String q = null;
         ConnectionClass obj= new ConnectionClass();
         if (account.equals("Admin"))
         {
            q="select username from teacher";
         }
         else if(account.equals("Teacher"))
         {
            q="select username from teacher where username='"+username+"'";
```

```
    }
    ResultSet rest=obj.stm.executeQuery(q);
    while(rest.next())
    {
        ch1.add(rest.getString("username"));
    }
}
catch(Exception ex)
{
    ex.printStackTrace();
}

l1 = new JLabel("Update Teacher Details");
l1.setHorizontalAlignment(JLabel.CENTER);
l1.setForeground(new Color(176, 4, 21));
l1.setFont(f);

l2 = new JLabel("Username");
l3 = new JLabel("Name");
l4 = new JLabel("Email");
l5 = new JLabel("Father Name");
l6 = new JLabel("Phone");
l7 = new JLabel("City");
l8 = new JLabel("Gender");
l9 = new JLabel("Blood");
l10 = new JLabel("Age");
l11 = new JLabel("DOB");
l12 = new JLabel("Experience");

l2.setFont(f1);
l3.setFont(f1);
l4.setFont(f1);
l5.setFont(f1);
l6.setFont(f1);
l7.setFont(f1);
l8.setFont(f1);
l9.setFont(f1);
l10.setFont(f1);
l11.setFont(f1);
l12.setFont(f1);

l2.setForeground(Color.DARK_GRAY);

tf1=new JTextField();
```

```
tf1.setFont(f1);
tf2=new JTextField();
tf2.setFont(f1);
tf3=new JTextField();
tf3.setFont(f1);
tf4=new JTextField();
tf4.setFont(f1);
tf5=new JTextField();
tf5.setFont(f1);
tf6=new JTextField();
tf6.setFont(f1);
tf7=new JTextField();
tf7.setFont(f1);
tf8=new JTextField();
tf8.setFont(f1);
tf9=new JTextField();
tf9.setFont(f1);
tf10=new JTextField();
tf10.setFont(f1);

bt1=new JButton("Update Teacher");
bt1.setFont(f1);
bt1.setForeground(Color.WHITE);
bt1.setBackground(Color.RED);

bt2=new JButton("Back");
bt2.setFont(f1);
bt2.setForeground(Color.WHITE);
bt2.setBackground(Color.BLACK);


bt1.addActionListener(this);
bt2.addActionListener(this);

p1=new JPanel();
p1.setLayout(new GridLayout(1, 1, 10, 10));
p1.add(l1);

p2 = new JPanel();
p2.setLayout(new GridLayout(12, 2, 10, 10));
p2.add(l2);
p2.add(ch1);
p2.add(l3);
p2.add(tf1);
```

```
        p2.add(l4);
        p2.add(tf2);
        p2.add(l5);
        p2.add(tf3);
        p2.add(l6);
        p2.add(tf4);
        p2.add(l7);
        p2.add(tf5);
        p2.add(l8);
        p2.add(tf6);
        p2.add(l9);
        p2.add(tf7);
        p2.add(l10);
        p2.add(tf8);
        p2.add(l11);
        p2.add(tf9);
        p2.add(l12);
        p2.add(tf10);

        p3=new JPanel();
        p3.add(bt1);
        p3.add(bt2);

        setLayout(new BorderLayout(10, 10));
        add(p1, "North");
        add(p2, "Center");
        add(p3, "South");

        p2.setBorder(BorderFactory.createEmptyBorder(20, 40, 20, 40));
        p3.setBorder(BorderFactory.createEmptyBorder(10, 10, 10, 10));

        ch1.addItemListener(new ItemListener()
        {
            @Override
            public void itemStateChanged(ItemEvent e)
            {
                if (e.getStateChange() == ItemEvent.SELECTED)
                {
                    try
                    {
                        ConnectionClass obj= new ConnectionClass();
                        String username=ch1.getSelectedItem();
                        String q = "select name, email, father_name, phone, city, gender, blood_group,
age, dob, teaching_exp from teacher where username='" + username + "'";
```

```
                ResultSet rest=obj.stm.executeQuery(q);
                while (rest.next())
                {
                    tf1.setText(rest.getString("name"));
                    tf2.setText(rest.getString("email"));
                    tf3.setText(rest.getString("father_name"));
                    tf4.setText(rest.getString("phone"));
                    tf5.setText(rest.getString("city"));
                    tf6.setText(rest.getString("gender"));
                    tf7.setText(rest.getString("blood_group"));
                    tf8.setText(rest.getString("age"));
                    tf9.setText(rest.getString("dob"));
                    tf10.setText(rest.getString("teaching_exp"));
                }
            }
            catch(Exception ex)
            {
                ex.printStackTrace();
            }
        }
    }
});

}

@Override
public void actionPerformed(ActionEvent ae)
{
    if(ae.getSource()==bt1)
    {
        String username=ch1.getSelectedItem();
        String name=tf1.getText();
        String email=tf2.getText();
        String father_name=tf3.getText();
        String phone=tf4.getText();
        String city=tf5.getText();
        String gender=tf6.getText();
        String blood_group=tf7.getText();
        String age=tf8.getText();
        String dob=tf9.getText();
        String exp=tf10.getText();

        try
        {
```

```
            ConnectionClass obj=new ConnectionClass();
            String q = "update teacher set name='"+name+"', email='"+email+"',
father_name='"+father_name+"', phone='"+phone+"', city='"+city+"', gender='"+gender+"',
blood_group='"+blood_group+"', age='"+age+"', dob='"+dob+"', teaching_exp='"+exp+"' where
username='"+username+"'";
            obj.stm.executeUpdate(q);
            int result = obj.stm.executeUpdate(q);
            if (result == 1)
            {
               JOptionPane.showMessageDialog(null, "Class Details Successfully Updated");
               this.setVisible(false);
            }
            else
            {
               JOptionPane.showMessageDialog(null, "Please Fill all the Details");
               this.setVisible(false);
               new UpdateClassDetails().setVisible(true);
            }
            setVisible(false);
         }
         catch(Exception ex)
         {
            ex.printStackTrace();
         }
      }
      if(ae.getSource()==bt2)
      {
         setVisible(false);
      }
   }
//    public static void main(String[] args)
//    {
//        new UpdateTeacherDetails("account2", "pubu").setVisible(true);
//    }
}
```

ViewTeacherDetails.java

```
package School_Management_System;

import java.awt.event.*;
import java.awt.*;
import javax.swing.*;
import java.sql.*;

public class ViewTeacherDetails extends JFrame implements ActionListener
{
   String x[] = {"Name", "Username", "Email", "Phone", "Blood Group", "Gender", "City", "Age",
"DOB"};
   String y[][] = new String[30][9];
   int i=0, j=0;
   JTable t;
   Font f, f1;
   JLabel l1, l2;
   JButton bt1;
   JPanel p1, p2, p3;
   Choice ch1;

   ViewTeacherDetails(String account, String username)
   {
      super("Teacher Information");
      setSize(1500,550);
      setResizable(false);
      setLocation(200,200);
      f= new Font("Arial",Font.BOLD,15);
      try
      {
         ConnectionClass obj=new ConnectionClass();
         String q;
         if (account.equals("Admin")) {
            q = "SELECT * FROM teacher";
         } else {
            q = "SELECT * FROM teacher WHERE username='" + username + "'";
         }
         ResultSet rest = obj.stm.executeQuery(q);
         while(rest.next())
         {
            y[i][j++]=rest.getString("name");
            y[i][j++] = rest.getString("username");
            y[i][j++]=rest.getString("email");
            y[i][j++]=rest.getString("phone");
```

```
            y[i][j++]=rest.getString("blood_group");
            y[i][j++]=rest.getString("gender");
            y[i][j++]=rest.getString("city");
            y[i][j++]=rest.getString("age");
            y[i][j++]=rest.getString("dob");
            i++;
            j=0;
        }
        t= new JTable(y,x);
        t.setFont(f);
        t.setBackground(Color.BLACK);
        t.setForeground(Color.WHITE);
        JScrollPane sp=new JScrollPane(t);
        if (account.equals("Admin"))
        {
            f1=new Font("Lucida Fax",Font.BOLD,25);

            l1=new JLabel("Delete any Teacher ?");
            l1.setForeground(Color.YELLOW);
            l1.setFont(f1);

            l2=new JLabel("Teacher Username");
            l2.setBackground(Color.BLACK);
            l2.setForeground(Color.GRAY);
            l2.setFont(f1);

            ch1 = new Choice();
            ch1.add("Select Username");
            ch1.setSize(100, 100);
            ch1.setFont(f1);
            ConnectionClass obj2 = new ConnectionClass();
            ResultSet rs2 = obj2.stm.executeQuery("SELECT username FROM teacher");

            while (rs2.next()) {
                ch1.add(rs2.getString("username"));
            }

            bt1=new JButton("Delete Teacher");
            bt1.addActionListener(this);
            bt1.setBackground(Color.BLACK);
            bt1.setForeground(Color.RED);
            bt1.setFont(f1);

            p1=new JPanel();
```

```java
            p1.setLayout(new GridLayout(1, 1, 10, 10));
            p1.add(l1);

            p2=new JPanel();
            p2.setLayout(new GridLayout(1, 3, 10, 10));
            p2.add(l2);
            p2.add(ch1);
            p2.add(bt1);

            p3=new JPanel();
            p3.setLayout(new GridLayout(2, 1, 10, 10));
            p3.add(p1);
            p3.add(p2);

            p1.setBackground(Color.BLACK);
            p2.setBackground(Color.BLACK);
            p3.setBackground(Color.BLACK);
            add(p3, "South");
        }
        add (sp);
    }
    catch(Exception ex)
    {
        ex.printStackTrace();
    }
}
public void actionPerformed(ActionEvent e)
{
    if (e.getSource() == bt1)
    {
        String username = ch1.getSelectedItem();

        // ✅ CONFIRMATION MUST COME HERE
        int confirm = JOptionPane.showConfirmDialog(
                this,
                "Are you sure you want to delete teacher: " + username + "?",
                "Confirm Delete",
                JOptionPane.YES_NO_OPTION
        );

        if (confirm != JOptionPane.YES_OPTION)
        {
            return; // stop deletion
        }
```

```
            try
            {
                ConnectionClass obj = new ConnectionClass();
                String q = "DELETE FROM teacher WHERE username='" + username + "'";
                int res = obj.stm.executeUpdate(q);

                if (res == 1)
                {
                    JOptionPane.showMessageDialog(this, "Teacher record is deleted!");
                    setVisible(false);
                }
                else
                {
                    JOptionPane.showMessageDialog(this, "Teacher record does not match!");
                }
            }
            catch (Exception ex)
            {
                ex.printStackTrace();
            }
        }
    }

//    public static void main(String[] args)
//    {
//        new ViewTeacherDetails("account", "username").setVisible(true);
//    }
}
```

AddStudentDetails.java

```
package School_Management_System;

import java.awt.*;
import javax.swing.*;
import java.awt.event.*;
import java.security.*;
import java.sql.*;
import java.util.Random;

public class AddStudentDetails extends JFrame implements ActionListener
{
   JLabel l1, l2, l3, l4, l5, l6, l7, l8, l9, l10, l11, l12, l13, l14;
   JButton bt1, bt2;
   JTextField tf1, tf2, tf3, tf4, tf5, tf6, tf7, tf8, tf9, tf10;
   Choice ch1;
   JComboBox ch2;
   JPasswordField pf1;
   String class_name=null;

   AddStudentDetails()
   {
      setTitle("Add New Student Details");
      getContentPane().setBackground(Color.WHITE);
      setLayout(null);
      setSize(840, 600);
      setResizable(false);
      setLocation(550,200);
      setVisible(true);

      // Load background
      ImageIcon img = new ImageIcon(getClass().getResource("Icons/studentadd.jpg"));
      Image scaledImg = img.getImage().getScaledInstance(840, 600,
Image.SCALE_SMOOTH);
      JLabel bg1 = new JLabel(new ImageIcon(scaledImg));
      bg1.setBounds(0, 0, 840, 600);
      bg1.setLayout(null);
      add(bg1);

      l1= new JLabel("Add Student Details for Signup");
      l1.setBounds(230, 50, 500, 50);
      l1.setFont(new Font("Ariel", Font.BOLD, 30));
      l1.setForeground(Color.BLACK);
      bg1.add(l1);
```

```
l2= new JLabel("Name");
l2.setBounds(50, 150, 150, 30);
l2.setFont(new Font("Ariel", Font.BOLD, 20));
l2.setForeground(Color.WHITE);
bg1.add(l2);

tf1= new JTextField();
tf1.setBounds(200, 150, 150, 30);
bg1.add(tf1);

l3= new JLabel("Username");
l3.setBounds(450, 150, 150, 30);
l3.setFont(new Font("Ariel", Font.BOLD, 20));
l3.setForeground(Color.WHITE);
bg1.add(l3);

tf2= new JTextField();
tf2.setBounds(600, 150, 150, 30);
bg1.add(tf2);

l4= new JLabel("Password");
l4.setBounds(50, 200, 150, 30);
l4.setFont(new Font("Ariel", Font.BOLD, 20));
l4.setForeground(Color.WHITE);
bg1.add(l4);

pf1= new JPasswordField();
pf1.setBounds(200, 200, 150, 30);
bg1.add(pf1);

l5= new JLabel("E-Mail");
l5.setBounds(450, 200, 150, 30);
l5.setFont(new Font("Ariel", Font.BOLD, 20));
l5.setForeground(Color.WHITE);
bg1.add(l5);

tf3= new JTextField();
tf3.setBounds(600, 200, 150, 30);
bg1.add(tf3);

l6= new JLabel("Father's name");
l6.setBounds(50, 250, 150, 30);
l6.setFont(new Font("Ariel", Font.BOLD, 20));
```

```
l6.setForeground(Color.WHITE);
bg1.add(l6);

tf4= new JTextField();
tf4.setBounds(200, 250, 150, 30);
bg1.add(tf4);

l7= new JLabel("Phone");
l7.setBounds(450, 250, 150, 30);
l7.setFont(new Font("Ariel", Font.BOLD, 20));
l7.setForeground(Color.WHITE);
bg1.add(l7);

tf5= new JTextField();
tf5.setBounds(600, 250, 150, 30);
bg1.add(tf5);

l8= new JLabel("Blood Group");
l8.setBounds(50, 300, 150, 30);
l8.setFont(new Font("Ariel", Font.BOLD, 20));
l8.setForeground(Color.WHITE);
bg1.add(l8);

tf6= new JTextField();
tf6.setBounds(200, 300, 150, 30);
bg1.add(tf6);

l9= new JLabel("Gender");
l9.setBounds(450, 300, 150, 30);
l9.setFont(new Font("Ariel", Font.BOLD, 20));
l9.setForeground(Color.WHITE);
bg1.add(l9);


ch1=new Choice();
ch1.add("Select Gender");
ch1.add("Male");
ch1.add("Female");
ch1.setFont(new Font("Arial", Font.BOLD, 18));
ch1.setBounds(600, 300, 150, 30);
bg1.add(ch1);

// City
l10 = new JLabel("City");
```

```java
l10.setBounds(50, 350, 150, 30);
l10.setFont(new Font("Ariel", Font.BOLD, 20));
l10.setForeground(Color.WHITE);
bg1.add(l10);

tf7 = new JTextField();
tf7.setBounds(200, 350, 150, 30);
bg1.add(tf7);

// AGE
l11 = new JLabel("Age");
l11.setBounds(450, 350, 150, 30);
l11.setFont(new Font("Ariel", Font.BOLD, 20));
l11.setForeground(Color.WHITE);
bg1.add(l11);

tf8 = new JTextField();
tf8.setBounds(600, 350, 150, 30);
bg1.add(tf8);

// GRADE
l12 = new JLabel("Grade");
l12.setBounds(50, 400, 200, 30);
l12.setFont(new Font("Ariel", Font.BOLD, 20));
l12.setForeground(Color.WHITE);
bg1.add(l12);

// create JComboBox
JComboBox<String> ch2 = new JComboBox<>();
ch2.addItem("Select Grade");
try {
   ConnectionClass obj = new ConnectionClass();
   String q = "select distinct * from class";
   ResultSet rest = obj.stm.executeQuery(q);
   while (rest.next()) {
      ch2.addItem(rest.getString("class_name"));
   }
   rest.close();
} catch (Exception ex) {
   ex.printStackTrace();
}

// size, font, and placement
ch2.setBounds(200, 400, 150, 30);
```

```
        ch2.setFont(new Font("Ariel", Font.BOLD, 14));
        ch2.setFocusable(false);            // optional
        ((JComponent)ch2.getRenderer()).setOpaque(false); // keep renderer transparent if
needed
        bg1.add(ch2);

        // DOB (fixed position)
        l13 = new JLabel("DOB");
        l13.setBounds(450, 400, 150, 30);  // moved down from 350 → 400
        l13.setFont(new Font("Ariel", Font.BOLD, 20));
        l13.setForeground(Color.WHITE);
        bg1.add(l13);

        tf9 = new JTextField();
        tf9.setBounds(600, 400, 150, 30);  // aligned with DOB
        bg1.add(tf9);

        // ROLL NO
        l14 = new JLabel("Roll No");
        l14.setBounds(250, 450, 150, 30);   // stays fine
        l14.setFont(new Font("Ariel", Font.BOLD, 20));
        l14.setForeground(Color.BLACK);
        bg1.add(l14);

        tf10 = new JTextField();
        tf10.setBounds(400, 450, 150, 30);
        tf10.setEditable(false);
        bg1.add(tf10);

        //buttons
        bt1= new JButton("Submit");
        bt1.setBackground(Color.BLACK);
        bt1.setForeground(Color.WHITE);
        bt1.setBounds(250, 500, 150, 30);

        bt2= new JButton("Cancel");
        bt2.setBackground(new Color(8, 161, 92));
        bt2.setForeground(Color.BLACK);
        bt2.setBounds(500, 500, 150, 30);

        bg1.add(bt1);
        bg1.add(bt2);

        bt1.addActionListener(this);
```

```
bt2.addActionListener(this);

/*ch2.addMouseListener(new MouseAdapter()
{
   @Override
   public void mouseClicked(MouseEvent e)
   {
      try
      {
         ConnectionClass obj=new ConnectionClass();
         String class_name = (String) ch2.getSelectedItem();
         String q1="select * from class where class_name='"+class_name+"'";
         ResultSet rest1=obj.stm.executeQuery(q1);
         while (rest1.next())
         {
            int roll_no=Integer.parseInt(rest1.getString("enrolled"))+1;
            tf10.setText(String.valueOf(roll_no));
         }
      }
      catch(Exception ex)
      {
         ex.printStackTrace();
      }
   }
}); */

ch2.addActionListener(new ActionListener() {
@Override
public void actionPerformed(ActionEvent e) {
   try {
      // get selected value from combobox
      class_name = (String) ch2.getSelectedItem();

      // show the selected class_name (example: print or set in a text field)
      System.out.println("Selected class: " + class_name);

      ConnectionClass obj = new ConnectionClass();
      String q1 = "select * from class where class_name='" + class_name + "'";
      ResultSet rest1 = obj.stm.executeQuery(q1);

      while (rest1.next()) {
         int roll_no = Integer.parseInt(rest1.getString("enrolled")) + 1;
         tf10.setText(String.valueOf(roll_no));
      }
```

```
            rest1.close();
        } catch (Exception ex) {
            ex.printStackTrace();
        }
    }
});
//Force UI refresh and visibility
revalidate();
repaint();
setVisible(true);
}


@Override
public void actionPerformed(ActionEvent e)
{
    if(e.getSource()==bt1)
    {
        String name=tf1.getText();
        String roll_no=tf10.getText();
        String username=tf2.getText();
        String password=pf1.getText();
        // Hash the password with a salt
        String hashedPassword=null;
            try {
                hashedPassword = PasswordUtils.hashPassword(password);
            }
            catch (NoSuchAlgorithmException ex) { }
                //Logger.getLogger(UserScreen.class.getName()).log(Level.SEVERE, null, ex);
        String email=tf3.getText();
        String father_name=tf4.getText();
        String phone=tf5.getText();
        String blood_group=tf6.getText();
        String gender=(String) ch1.getSelectedItem();
        String city=tf7.getText();
        String age=tf8.getText();
        String grade=class_name;
        String dob=tf9.getText();
        Random r= new Random();
        String stu_id=""+Math.abs(r.nextInt() % 100000);
        int total_student=0;

        try
```

```
            {
                ConnectionClass obj= new ConnectionClass();
                String q= "insert into student
values('"+stu_id+"','"+roll_no+"','"+name+"','"+username+"','"+hashedPassword+"','"+email+"','"+f
ather_name+"','"+phone+"','"+blood_group+"','"+gender+"','"+city+"','"+age+"','"+grade+"','"+dob+
"')";
                String q1="update class set enrolled='"+roll_no+"' where class_name='"+grade+"'";
                String q2="select * from class where class_name='"+grade+"'";

                ResultSet rest=obj.stm.executeQuery(q2);
                while(rest.next())
                {
                    total_student=Integer.parseInt(rest.getString("strength"));
                }
                if (Integer.parseInt(roll_no)>total_student)
                {
                    JOptionPane.showMessageDialog(null,"Sorry ! This class is full");
                }
                else
                {
                    obj.stm.executeUpdate(q);
                    obj.stm.executeUpdate(q1);
                    JOptionPane.showMessageDialog(null,"Details Successfully Inserted");
                    setVisible(false);
                }
            }
            catch(Exception ex)
            {
                ex.printStackTrace();
            }
        }
        if (e.getSource()==bt2)
        {
            setVisible(false);
        }
    }

//    public static void main(String[] args)
//    {
//        new AddStudentDetails().setVisible(true);
//    }
}
```

UpdateStudentDetails.java

```
package School_Management_System;

import java.awt.*;
import javax.swing.*;
import java.awt.event.*;
import java.sql.*;

public class UpdateStudentDetails extends JFrame implements ActionListener
{
    JLabel l1, l2, l3, l4, l5, l6, l7, l8, l9, l10, l11, l12, l13, l14;
    JPanel p1, p2, p3;
    JTextField tf1, tf2, tf3, tf4, tf5, tf6, tf7, tf8, tf9, tf10, tf11;
    Choice ch1;
    JButton bt1, bt2;
    public String pubu, account2;

    UpdateStudentDetails(String account, String username)
    {
        super("Update Student Details");
        setSize(760, 720);
        setLocation(550,200);
        setVisible(true);
        account2=account;
        pubu=username;

        Font f = new Font("Arial", Font.BOLD, 28);
        Font f1 = new Font("Arial", Font.PLAIN, 18);

        ch1=new Choice();
        ch1.setFont(f1);
        ch1.add("Select Username");

        try
        {
            ConnectionClass obj= new ConnectionClass();
            String q;
            q="";
            if (account.equals("Admin"))
            {
                q="select username from student";
            }
            else if (account.equals("Teacher"))
            {
```

```
        q="select username from student";
    }
    else if(account.equals("Student"))
    {
        q="select username from student where username='"+username+"'";
    }
    ResultSet rest=obj.stm.executeQuery(q);
    while(rest.next())
    {
        ch1.add(rest.getString("username"));
    }
}
catch(Exception ex)
{
    ex.printStackTrace();
}

l1 = new JLabel("Update Student Details");
l1.setHorizontalAlignment(JLabel.CENTER);
l1.setForeground(new Color(176, 4, 21));
l1.setFont(f);

l2 = new JLabel("Username");
l3 = new JLabel("Roll No");
l4 = new JLabel("Name");
l5 = new JLabel("Email");
l6 = new JLabel("Father Name");
l7 = new JLabel("Phone");
l8 = new JLabel("Class");
l9 = new JLabel("Gender");
l10 = new JLabel("Blood");
l11 = new JLabel("Age");
l12 = new JLabel("DOB");
l13 = new JLabel("City");


l2.setFont(f1);
l3.setFont(f1);
l4.setFont(f1);
l5.setFont(f1);
l6.setFont(f1);
l7.setFont(f1);
l8.setFont(f1);
l9.setFont(f1);
```

```
l10.setFont(f1);
l11.setFont(f1);
l12.setFont(f1);
l13.setFont(f1);

l2.setForeground(Color.DARK_GRAY);

tf1=new JTextField();
tf1.setFont(f1);
tf1.setEditable(false);
tf2=new JTextField();
tf2.setFont(f1);
tf3=new JTextField();
tf3.setEditable(false);
tf3.setFont(f1);
tf4=new JTextField();
tf4.setFont(f1);
tf5=new JTextField();
tf5.setFont(f1);
tf6=new JTextField();
tf6.setFont(f1);
tf7=new JTextField();
tf7.setFont(f1);
tf8=new JTextField();
tf8.setFont(f1);
tf9=new JTextField();
tf9.setFont(f1);
tf10=new JTextField();
tf10.setFont(f1);
tf11=new JTextField();
tf11.setFont(f1);

bt1=new JButton("Update Student");
bt1.setFont(f1);
bt1.setForeground(Color.WHITE);
bt1.setBackground(Color.RED);

bt2=new JButton("Back");
bt2.setFont(f1);
bt2.setForeground(Color.WHITE);
bt2.setBackground(Color.BLACK);


bt1.addActionListener(this);
```

```java
bt2.addActionListener(this);

p1=new JPanel();
p1.setLayout(new GridLayout(1, 1, 10, 10));
p1.add(l1);

p2 = new JPanel();
p2.setLayout(new GridLayout(12, 2, 10, 10));
p2.add(l2);
p2.add(ch1);
p2.add(l3);
p2.add(tf1);
p2.add(l4);
p2.add(tf2);
p2.add(l5);
p2.add(tf3);
p2.add(l6);
p2.add(tf4);
p2.add(l7);
p2.add(tf5);
p2.add(l8);
p2.add(tf6);
p2.add(l9);
p2.add(tf7);
p2.add(l10);
p2.add(tf8);
p2.add(l11);
p2.add(tf9);
p2.add(l12);
p2.add(tf10);

p3=new JPanel();
p3.add(bt1);
p3.add(bt2);

setLayout(new BorderLayout(10, 10));
add(p1, "North");
add(p2, "Center");
add(p3, "South");

p2.setBorder(BorderFactory.createEmptyBorder(20, 40, 20, 40));
p3.setBorder(BorderFactory.createEmptyBorder(10, 10, 10, 10));

ch1.addItemListener(new ItemListener()
```

```
        {
            @Override
            public void itemStateChanged(ItemEvent e)
            {
                if (e.getStateChange() == ItemEvent.SELECTED)
                {
                    try
                    {
                        ConnectionClass obj= new ConnectionClass();
                        String username=ch1.getSelectedItem();
                        String q = "select roll_no, name, email, father_name, phone, city, gender, blood,
age, dob, class from student where username='" + username + "'";
                        ResultSet rest=obj.stm.executeQuery(q);
                        while (rest.next())
                        {
                            tf1.setText(rest.getString("roll_no"));
                            tf2.setText(rest.getString("name"));
                            tf3.setText(rest.getString("email"));
                            tf4.setText(rest.getString("father_name"));
                            tf5.setText(rest.getString("phone"));
                            tf6.setText(rest.getString("class"));
                            tf7.setText(rest.getString("gender"));
                            tf8.setText(rest.getString("blood"));
                            tf9.setText(rest.getString("age"));
                            tf10.setText(rest.getString("dob"));
                            tf11.setText(rest.getString("city"));
                        }
                    }
                    catch(Exception ex)
                    {
                        ex.printStackTrace();
                    }
                }
            }
        });

    }

    @Override
    public void actionPerformed(ActionEvent ae)
    {
            if(ae.getSource()==bt1)
        {
            String username=ch1.getSelectedItem();
```

```
            String name=tf2.getText();
            String email=tf3.getText();
            String father_name=tf4.getText();
            String phone=tf5.getText();
            String city=tf11.getText();
            String gender=tf7.getText();
            String blood_group=tf8.getText();
            String age=tf9.getText();
            String dob=tf10.getText();


            try
            {
                ConnectionClass obj=new ConnectionClass();
                String q = "update student set name='"+name+"', email='"+email+"',
father_name='"+father_name+"', phone='"+phone+"', city='"+city+"', gender='"+gender+"',
blood='"+blood_group+"', age='"+age+"', dob='"+dob+"' where username='"+username+"'";
                obj.stm.executeUpdate(q);
                int result = obj.stm.executeUpdate(q);
                if (result == 1)
                {
                    JOptionPane.showMessageDialog(null, "Class Details Successfully Updated");
                    this.setVisible(false);
                }
                else
                {
                    JOptionPane.showMessageDialog(null, "Please Fill all the Details");
                    this.setVisible(false);
                    new UpdateClassDetails().setVisible(true);
                }
                setVisible(false);
            }
            catch(Exception ex)
            {
                ex.printStackTrace();
            }
        }
        if(ae.getSource()==bt2)
        {
            setVisible(false);
        }
    }
//    public static void main(String[] args)
//    {
```

```
//        new UpdateStudentDetails("account2", "pubu").setVisible(true);
//    }
}
```

ViewStudentDetails.java

```
package School_Management_System;

import java.awt.*;
import javax.swing.*;
import java.awt.event.*;
import java.sql.*;

public class ViewStudentDetails extends JFrame implements ActionListener
{
   String x[]={"Roll No", "Name", "Username", "Email", "Father Name", "Phone", "Blood",
"Gender", "City", "Age", "Class", "DOB"};
   String y[][]=new String[30][12];
   int i=0,j=0;
   JTable t;
   Font f;
   String query;
   public static String pu, pa;
   Choice ch1;
   JButton bt1;
   JPanel p1, p2, p3;
   JLabel l1, l2;
   Font f1;

   ViewStudentDetails(String pub_username, String account2)
   {
      super("Student Details");
      setSize(1500, 500);
      setResizable(true);
      setLocation(1,1);
      f=new Font("MS UI Gothic", Font.BOLD, 15);

      pa=pub_username;
      pu=account2;

      System.out.println(pu +  "pub_username" + "Account type :" + pa);
      try
      {
         ConnectionClass obj=new ConnectionClass();
         if(pa.equals("Student"))
         {
            query="select * from student where username='"+pu+"'";
         }
         else if(pa.equals("Admin"))
```

```
{
   query="select * from student";
}
else if(pa.equals("Teacher"))
{
   query="select * from student";
}
ResultSet rest=obj.stm.executeQuery(query);
while(rest.next())
{
   y[i][j++]=rest.getString("roll_no");
   y[i][j++]=rest.getString("name");
   y[i][j++]=rest.getString("username");
   y[i][j++]=rest.getString("email");
   y[i][j++]=rest.getString("father_name");
   y[i][j++]=rest.getString("phone");
   y[i][j++]=rest.getString("blood");
   y[i][j++]=rest.getString("gender");
   y[i][j++]=rest.getString("city");
   y[i][j++]=rest.getString("age");
   y[i][j++]=rest.getString("class");
   y[i][j++]=rest.getString("dob");
   i++;
   j=0;
}

t=new JTable(y,x);
t.setFont(f);
t.setBackground(Color.BLACK);
t.setForeground(Color.WHITE);
if (pa.equals("Admin"))
{
   f1 = new Font("Lucida Fax", Font.BOLD, 20);

   l1 = new JLabel("Delete any Student ?");
   l1.setForeground(Color.YELLOW);
   l1.setFont(f1);

   l2 = new JLabel("Student Username");
   l2.setForeground(Color.GRAY);
   l2.setFont(f1);

   ch1 = new Choice();
   ch1.add("Select Username");
```

```
            ch1.setFont(f1);

            ConnectionClass obj2 = new ConnectionClass();
            ResultSet rs2 = obj2.stm.executeQuery("SELECT username FROM student");

            while (rs2.next())
            {
              ch1.add(rs2.getString("username"));
            }

            bt1 = new JButton("Delete Student");
            bt1.setFont(f1);
            bt1.setForeground(Color.RED);
            bt1.setBackground(Color.BLACK);
            bt1.addActionListener(this);

            p1 = new JPanel(new GridLayout(1,1));
            p1.add(l1);

            p2 = new JPanel(new GridLayout(1,3,10,10));
            p2.add(l2);
            p2.add(ch1);
            p2.add(bt1);

            p3 = new JPanel(new GridLayout(2,1));
            p3.add(p1);
            p3.add(p2);

            p1.setBackground(Color.BLACK);
            p2.setBackground(Color.BLACK);
            p3.setBackground(Color.BLACK);

            add(p3, BorderLayout.SOUTH);
          }
        }
        catch(Exception ex)
        {
          ex.printStackTrace();
        }
        JScrollPane sp = new JScrollPane(t);
        add(sp);
    }
    public void actionPerformed(ActionEvent e)
        {
```

```
        if (e.getSource() == bt1)
        {
            String username = ch1.getSelectedItem();

            if (username.equals("Select Username"))
            {
                JOptionPane.showMessageDialog(this, "Please select a student");
                return;
            }

            int confirm = JOptionPane.showConfirmDialog(this, "Are you sure you want to delete
student: " + username + "?", "Confirm Delete", JOptionPane.YES_NO_OPTION);

            if (confirm != JOptionPane.YES_OPTION)
                return;

            try
            {
                ConnectionClass obj = new ConnectionClass();
                String q = "DELETE FROM student WHERE username='" + username + "'";
                int res = obj.stm.executeUpdate(q);

                if (res == 1)
                {
                    JOptionPane.showMessageDialog(this, "Student record deleted!");
                    setVisible(false);
                    new ViewStudentDetails(pu, pa).setVisible(true);
                }
                else
                {
                    JOptionPane.showMessageDialog(this, "Student not found!");
                }
            }
            catch (Exception ex)
            {
                ex.printStackTrace();
            }
        }
    }

//    public static void main(String[] args)
//    {
//        new ViewStudentDetails(pu, pa).setVisible(true);
//    }
```

```
}
```

AddNewClass.java

```
package School_Management_System;

import java.awt.*;
import javax.swing.*;
import java.awt.event.*;
import java.sql.*;
import java.util.Random;

public class AddNewClass extends JFrame implements ActionListener
{
   JLabel l1, l2, l3, l4, l5;
   JTextField tf1;
   Choice ch1, ch2, ch3;
   JButton bt1, bt2;

   AddNewClass()
   {
      setTitle("Add New Class Details");
      getContentPane().setBackground(Color.WHITE);
      setLayout(null);
      setSize(840, 600);
      setResizable(false);
      setLocation(550,200);
      setVisible(true);

      // Load background
      ImageIcon img = new ImageIcon(getClass().getResource("Icons/classadd.jpg"));
      Image scaledImg = img.getImage().getScaledInstance(840, 600,
Image.SCALE_SMOOTH);
      JLabel bg1 = new JLabel(new ImageIcon(scaledImg));
      bg1.setBounds(0, 0, 840, 600);
      bg1.setLayout(null);
      add(bg1);

      l1= new JLabel("Add New Class Details");
      l1.setBounds(420, 110, 500, 50);
      l1.setFont(new Font("Ariel", Font.BOLD, 30));
      l1.setForeground(new Color(8, 161, 92));
      bg1.add(l1);

      l2=new JLabel("Class Name");
      l2.setBounds(400, 200, 150, 30);
      l2.setFont(new Font("Arial",Font.BOLD, 20));
```

```
l2.setForeground(Color.BLACK);
bg1.add(l2);

ch1= new Choice();
ch1.add("Select Grade");
ch1.add("VIII");
ch1.add("IX");
ch1.add("X");
ch1.add("IBDP-I");
ch1.add("IBDP-II");
ch1.add("AS-LEVEL");
ch1.add("A-LEVEL");
ch1.setBounds(600, 200, 150, 30);
ch1.setFont(new Font("Arial",Font.BOLD, 20));
bg1.add(ch1);

l2=new JLabel("Class Strength");
l2.setBounds(400, 250, 150, 30);
l2.setFont(new Font("Arial",Font.BOLD, 20));
l2.setForeground(Color.BLACK);
bg1.add(l2);

ch2= new Choice();
ch2.add("5");
ch2.add("10");
ch2.add("15");
ch2.add("20");
ch2.setBounds(600, 250, 150, 30);
ch2.setFont(new Font("Arial",Font.BOLD, 20));
bg1.add(ch2);

l3= new JLabel("Enrolled Students");
l3.setBounds(400, 300, 150, 30);
l3.setFont(new Font("Arial",Font.BOLD, 20));
l3.setForeground(Color.BLACK);
bg1.add(l3);

tf1= new JTextField();
tf1.setBounds(600, 300, 150, 30);
tf1.setFont(new Font("Arial",Font.BOLD, 20));
tf1.setEditable(false);
tf1.setText("0");
bg1.add(tf1);
```

```
        bt1=new JButton("Add Class");
        bt1.setBackground(Color.BLACK);
        bt1.setForeground(Color.WHITE);
        bt1.setBounds(400, 350, 150, 40);
        bg1.add(bt1);

        bt2=new JButton("Back");
        bt2.setBackground(new Color(88, 245, 174));
        bt2.setForeground(Color.BLACK);
        bt2.setBounds(600, 350, 150, 40);
        bg1.add(bt2);

        bt1.addActionListener(this);
        bt2.addActionListener(this);

        //Force UI refresh and visibility
        revalidate();
        repaint();
        setVisible(true);

    }
    public void actionPerformed(ActionEvent e)
    {
        if(e.getSource()==bt1)
        {
            String classname=ch1.getSelectedItem();
            String enrolled_students=ch2.getSelectedItem();
            String enrolled=tf1.getText();
            Random r= new Random();
            String cla_id=""+Math.abs(r.nextInt() % 100000);

            try
            {
                ConnectionClass obj=new ConnectionClass();
                String q="insert into class
values('"+cla_id+"','"+classname+"','"+enrolled_students+"','"+enrolled+"')";
                obj.stm.executeUpdate(q);
                JOptionPane.showMessageDialog(null, "Details Successfully Inserted");
                setVisible(false);
            }
            catch(Exception ex)
            {
                ex.printStackTrace();
            }
```

```
        }
        if(e.getSource()==bt2)
        {
            setVisible(false);
        }
    }
//    public static void main(String[] args)
//    {
//        new AddNewClass().setVisible(true);
//    }
}
```

UpdateClassDetails.java

```
package School_Management_System;

import java.awt.*;
import javax.swing.*;
import java.awt.event.*;
import java.sql.*;

public class UpdateClassDetails extends JFrame implements ActionListener
{
   JLabel l1, l2, l3, l4;
   JPanel p1, p2, p3;
   JTextField tf1, tf2;
   Choice ch1, ch2;
   JButton bt1, bt2;

   UpdateClassDetails()
   {

      setTitle("Update Class Details");
      getContentPane().setBackground(Color.WHITE);
      setLayout(null);
      setSize(700, 320);
      setResizable(false);
      setLocation(550,200);
      setVisible(true);

      Font f = new Font("Arial", Font.BOLD, 28);
      Font f1 = new Font("Arial", Font.PLAIN, 18);

      ch1= new Choice();
      ch1.add("Select Class");

      try
      {
         ConnectionClass obj= new ConnectionClass();
         String q="select distinct class_name from class";
         ResultSet rest=obj.stm.executeQuery(q);
         while(rest.next())
         {
            ch1.add(rest.getString("class_name"));
         }
      }
      catch(Exception ex)
```

```
{
   ex.printStackTrace();
}
l1=new JLabel("Update Class Details");
l1.setHorizontalAlignment(JLabel.CENTER);
l1.setForeground(new Color(176, 4, 21));
l1.setFont(f);


l2=new JLabel("Class Name");
l2.setFont(f1);

l3=new JLabel("Class Strength");
l3.setFont(f1);

l4=new JLabel("Enrolled Students");
l4.setFont(f1);

tf1=new JTextField();
tf1.setFont(f1);

tf2=new JTextField();
tf2.setFont(f1);
tf2.setEditable(false);

bt1=new JButton("Update");
bt1.setFont(f1);
bt1.setForeground(Color.WHITE);
bt1.setBackground(Color.RED);

bt2=new JButton("Back");
bt2.setFont(f1);
bt2.setForeground(Color.WHITE);
bt2.setBackground(Color.BLACK);


bt1.addActionListener(this);
bt2.addActionListener(this);

p1=new JPanel();
p1.setLayout(new GridLayout(1, 1, 10, 10));
p1.add(l1);

p2=new JPanel();
```

```
        p2.setLayout(new GridLayout(3, 2, 10, 10));
        p2.add(l2);
        p2.add(ch1);
        p2.add(l3);
        p2.add(tf1);
        p2.add(l4);
        p2.add(tf2);

        p3=new JPanel();
        p3.add(bt1);
        p3.add(bt2);

        setLayout(new BorderLayout(10, 10));
        add(p1, "North");
        add(p2, "Center");
        add(p3, "South");

        p2.setBorder(BorderFactory.createEmptyBorder(20, 40, 20, 40));
        p3.setBorder(BorderFactory.createEmptyBorder(10, 10, 10, 10));

        ch1.addMouseListener(new MouseAdapter()
        {
           @Override
           public void mouseClicked(MouseEvent e)
           {
              tf1.setText("");
              try
              {
                 ConnectionClass obj= new ConnectionClass();
                 String class_name=ch1.getSelectedItem();
                 String q = "select strength, enrolled from class where class_name='" + class_name
    + "'";
                 ResultSet rest=obj.stm.executeQuery(q);
                 while (rest.next())
                 {
                    tf1.setText(rest.getString("strength"));
                    tf2.setText(rest.getString("enrolled"));
                 }
              }
              catch(Exception ex)
              {
                 ex.printStackTrace();
              }
           }
```

```
        });
    }
    public void actionPerformed(ActionEvent ae)
    {
        if(ae.getSource()==bt1)
        {
            String classname=ch1.getSelectedItem();
            String enrolled=tf1.getText();
            String enrolled_students=tf2.getText();

            try
            {
                ConnectionClass obj=new ConnectionClass();
                String q = "update class set strength='" + enrolled + "' where class_name='" +
classname + "'";
                obj.stm.executeUpdate(q);
                int result = obj.stm.executeUpdate(q);
                if (result == 1)
                {
                    JOptionPane.showMessageDialog(null, "Class Details Successfully Updated");
                    this.setVisible(false);
                }
                else
                {
                    JOptionPane.showMessageDialog(null, "Please Fill all the Details");
                    this.setVisible(false);
                    new UpdateClassDetails().setVisible(true);
                }
                setVisible(false);
            }
            catch(Exception ex)
            {
                ex.printStackTrace();
            }
        }
        if(ae.getSource()==bt2)
        {
            setVisible(false);
        }
    }

//    public static void main(String[] args)
//    {
//        new UpdateClassDetails().setVisible(true);
```

```
//    }
}
```

AddSubjectDetails.java

```
package School_Management_System;

import java.awt.*;
import java.awt.event.*;
import java.sql.*;
import java.util.Random;
import javax.swing.*;

public class AddSubjectDetails extends JFrame implements ActionListener
{
    JLabel l1, l2, l3;
    JButton bt1, bt2;
    Choice ch1, ch2;

    AddSubjectDetails()
    {
        setTitle("Add New Subject Details");
        getContentPane().setBackground(Color.WHITE);

        setLocation(400,150);
        setSize(900, 600);

        ImageIcon img = new
ImageIcon(getClass().getResource("/School_Management_System/Icons/addsubject.jpg"));
        Image scaledImg = img.getImage().getScaledInstance(900, 600,
Image.SCALE_SMOOTH);
        JLabel bg1 = new JLabel(new ImageIcon(scaledImg));
        bg1.setBounds(0, 0, 900, 600);
        add(bg1);

        l1 = new JLabel("Add Subject Details", SwingConstants.CENTER);
        l1.setFont(new Font("Arial", Font.BOLD, 32));
        l1.setForeground(Color.BLACK);
        l1.setBounds(150, 80, 600, 50);
        bg1.add(l1);

        // Class Name
        l2 = new JLabel("Subject Code");
        l2.setFont(new Font("Arial", Font.BOLD, 20));
        l2.setForeground(Color.BLACK);
        l2.setBounds(250, 220, 200, 35);
        bg1.add(l2);
```

```
ch1=new Choice();
ch1.add("Select Subject Code Offered");
ch1.add("ECO");
ch1.add("BM");
ch1.add("GP");
ch1.setBounds(480, 220, 200, 35);
ch1.setFont(new Font("Arial", Font.BOLD, 22));
bg1.add(ch1);

l3=new JLabel("Subject Name");
l3.setBounds(250, 280, 200, 35);
l3.setFont(new Font("Arial",Font.BOLD,20));
l3.setForeground(Color.BLACK);
bg1.add(l3);

ch2=new Choice();
ch2.add("Economics");
ch2.add("Business Managment");
ch2.add("Global Politics");
ch2.setBounds(480, 280, 200, 35);
ch2.setFont(new Font("Arial", Font.BOLD, 22));
ch2.setEnabled(false);
bg1.add(ch2);

bt1=new JButton("Add Subject Details");
bt1.setBounds(200, 350, 200, 35);
bt1.setForeground(Color.WHITE);
bt1.setBackground(Color.BLACK);
bt1.addActionListener(this);
bg1.add(bt1);

bt2=new JButton("Back");
bt2.setBounds(480, 350, 200, 35);
bt2.setForeground(Color.BLACK);
bt2.setBackground(new Color(88,245,174));
bt2.addActionListener(this);
bg1.add(bt2);


ch1.addItemListener(new ItemListener()
{
   @Override
   public void itemStateChanged(ItemEvent e) {
```

```
            if (e.getStateChange() == ItemEvent.SELECTED) {

                ch2.removeAll(); // clear old items

                String code = ch1.getSelectedItem();

                if (code.equals("ECO")) {
                    ch2.add("Economics");
                }
                else if (code.equals("BM")) {
                    ch2.add("Business Management");
                }
                else if (code.equals("GP")) {
                    ch2.add("Global Politics");
                }
                else if(code.equals("Select Subject Code Offered")){
                    ch2.add("");
                }
            }
        }
    });
}
@Override
public void actionPerformed(ActionEvent e)
{
    if(e.getSource()==bt1)
    {
        Random r=new Random();
        String sub_id=""+Math.abs(r.nextInt() % 100000);
        String sub_code=ch1.getSelectedItem();
        String sub_name=ch2.getSelectedItem();

        if(sub_name == null || sub_name.isEmpty()){
            JOptionPane.showMessageDialog(null, "Select Subject Details Before Inserting!");
            return;
        }
        else if (sub_code==null||sub_code.isEmpty()){
            JOptionPane.showMessageDialog(null, "Select Proper Subject Details Before
Inserting!");
            return;
        }
        else if (sub_code != null && sub_name.isEmpty()==false);
        {
            try
```

```
                {
                    ConnectionClass obj=new ConnectionClass();
                    String q="insert into subject values('"+sub_id+"','"+sub_code+"', '"+sub_name+"')";
                    obj.stm.executeUpdate(q);
                    JOptionPane.showMessageDialog(null, "Subject Details Successfully Inserted");
                    setVisible(false);
                }
                catch(Exception ex)
                {
                    ex.printStackTrace();
                }
            }
        }
        else if(e.getSource()==bt2)
        {
            setVisible(false);
        }
    }
}

//    public static void main(String[] args)
//    {
//        new AddSubjectDetails().setVisible(true);
//    }
```

AddMarksDetails.java

```
package School_Management_System;

import java.awt.*;
import javax.swing.*;
import java.awt.event.*;
import java.sql.*;
import java.util.Random;

public class AddMarksDetails extends JFrame implements ActionListener
{

   JLabel l1, l2, l3, l4, l5, l6, l7;
   Choice ch1, ch2, ch3, ch4;
   JButton bt1, bt2;
   JTextField tf1, tf2;
   JPanel p1, p2;
   Font f, f1;

   AddMarksDetails()
   {
      super("Add Marks Detail");

      setSize(700, 480);
      setLocation(50, 10);
      setResizable(false);


      f=new Font("Arial", Font.BOLD,25);
      f1=new Font("Arial", Font.BOLD,18);

      ch1=new Choice();
      ch1.add("Select Grade");

      try
      {
         ConnectionClass obj=new ConnectionClass();
         String q="select distinct class from student";
         ResultSet rest=obj.stm.executeQuery(q);
         while(rest.next())
         {
            ch1.add(rest.getString("class"));
         }
      }
```

```
catch(Exception ex)
{
   ex.printStackTrace();
}

ch2=new Choice();
ch2.add("Select Username");
ch4=new Choice();
ch4.add("Select Subject");

try
{
   ConnectionClass obj=new ConnectionClass();
   String q="select distinct subject_name from subject";
   ResultSet rest=obj.stm.executeQuery(q);
   while(rest.next())
   {
      ch4.add(rest.getString("subject_name"));
   }
}
catch(Exception ex)
{
   ex.printStackTrace();
}

l1=new JLabel("Add Marks Details");
l1.setForeground(Color.RED);
l1.setHorizontalAlignment(JLabel.CENTER);

l2=new JLabel("Class Name");
l3=new JLabel("Student Username");
l4=new JLabel("Student Name");
l5=new JLabel("Subject");
l6=new JLabel("Marks");
l7=new JLabel("Term");

ch3=new Choice();
ch3.add("Select Term");
ch3.add("First Term");
ch3.add("Second Term");
ch3.add("Third Term");

tf1=new JTextField();
tf2=new JTextField();
```

```
bt1=new JButton("Add Marks");
bt1.setBackground(Color.RED);
bt1.setForeground(Color.WHITE);

bt2=new JButton("Back");
bt2.setBackground(Color.BLACK);
bt2.setForeground(Color.WHITE);

bt1.addActionListener(this);
bt2.addActionListener(this);

ch1.addItemListener(new ItemListener()
{
   @Override
   public void itemStateChanged(ItemEvent e) {
      if (e.getStateChange() == ItemEvent.SELECTED) {
         try {
            ConnectionClass obj = new ConnectionClass();
            String class_name = ch1.getSelectedItem();
            String q = "select username from student where class='" + class_name + "'";
            ResultSet rest = obj.stm.executeQuery(q);
            while (rest.next()) {
               ch2.add(rest.getString("username"));
            }
         } catch (Exception ex) {
            ex.printStackTrace();
         }
      }
   }
});

ch2.addItemListener(new ItemListener()
{
   @Override
   public void itemStateChanged(ItemEvent e) {
      if (e.getStateChange() == ItemEvent.SELECTED) {
         try {
            ConnectionClass obj = new ConnectionClass();
            String class_name = ch1.getSelectedItem();
            String username = ch2.getSelectedItem();
            String q = "select name from student where class='" + class_name + "' and
username='" + username + "'";
            ResultSet rest = obj.stm.executeQuery(q);
```

```
                if (rest.next()) {
                    tf1.setText(rest.getString("name"));
                }
            } catch (Exception ex) {
                ex.printStackTrace();
            }
        }
    }
});

p1=new JPanel();
p1.setLayout(new GridLayout(1,1,10,10));
p1.add(l1);

p2=new JPanel();
p2.setLayout(new GridLayout(8,2,10,10));
p2.add(l2);   // Class
p2.add(ch1);

p2.add(l3);   // Username
p2.add(ch2);

p2.add(l4);   // Student Name
p2.add(tf1);

p2.add(l5);   // Subject
p2.add(ch4);

p2.add(l6);   // Marks
p2.add(tf2);

p2.add(l7);   // Term
p2.add(ch3);

p2.add(bt1);  // Buttons
p2.add(bt2);

setLayout(new BorderLayout(10,10));
add(p1, "North");
add(p2, "Center");

l2.setFont(f);
l3.setFont(f);
l4.setFont(f);
```

```
    l5.setFont(f);
    l6.setFont(f);
    l7.setFont(f);

    ch1.setFont(f1);
    ch2.setFont(f1);
    ch3.setFont(f1);
    ch4.setFont(f1);
    tf1.setFont(f1);
    tf2.setFont(f1);

    bt1.setFont(f);
    bt2.setFont(f);


}
@Override
public void actionPerformed(ActionEvent e)
{
    if(e.getSource()==bt1)
    {
        String class_name=ch1.getSelectedItem();
        String name=tf1.getText();
        String username=ch2.getSelectedItem();
        String subject=ch4.getSelectedItem();
        String marks=tf2.getText();
        String term=ch3.getSelectedItem();
        Random r=new Random();
        String exam_id=""+Math.abs(r.nextInt() % 100000);

        if(marks.isEmpty())
        {
            JOptionPane.showMessageDialog(null, "Marks should not be empty");
        }
        else if(Integer.parseInt(marks)>500)
        {
            JOptionPane.showMessageDialog(null, "Marks should not be greater than 500");
        }
        else
        {
            try
            {
                ConnectionClass obj1=new ConnectionClass();
```

```
                String q = "insert into marks values('"+exam_id+"', '"+class_name+"',
'"+username+"', '"+name+"', '"+subject+"', '"+marks+"', '"+term+"')";
                obj1.stm.executeUpdate(q);
                JOptionPane.showMessageDialog(null,"Details Successfully Inserted");
                setVisible(false);
            }
            catch(Exception ee)
            {
                ee.printStackTrace();
            }
        }
    }
    if(e.getSource()==bt2)
    {
        setVisible(false);
    }
  }
//    public static void main(String[] args)
//    {
//        new AddMarksDetails().setVisible(true);
//    }
}
```

ViewMarksDetails.java

```
package School_Management_System;

import java.awt.*;
import javax.swing.*;
import java.awt.event.*;
import java.sql.*;

public class ViewMarksDetails extends JFrame
{
   String x[]={"Id", "Class Name", "Username", "Name", "Subject Name", "Marks", "Terms"};
   String y[][]=new String [21][7];
   int i=0,j=0;
   JTable t;
   String query;
   public static String pu, pa;
   Choice ch1;
   JButton bt1;
   JPanel p1, p2, p3;
   JLabel l1, l2;
   Font f, f1;

   ViewMarksDetails(String pub_username, String account2)
   {
      super("Marks Details");
      setSize(1500,400);
      setResizable(false);
      setLocation(1,1);

      f=new Font("Arial",Font.BOLD,15);

      pu=pub_username;
      pa=account2;
      String account= pa;

      try
      {
         ConnectionClass obj=new ConnectionClass();
         if(account.equals("Student"))
         {
            query="select * from marks where username='"+pu+"'";
         }
         else
         {
```

```
    query="select * from marks";
}
ResultSet rest=obj.stm.executeQuery(query);
while(rest.next())
{
    y[i][j++] = rest.getString("exam_id");
    y[i][j++] = rest.getString("class_name");
    y[i][j++] = rest.getString("username");
    y[i][j++] = rest.getString("name");
    y[i][j++] = rest.getString("subject_name");
    y[i][j++] = rest.getString("marks");
    y[i][j++] = rest.getString("term");
    i++;
    j=0;
}
t=new JTable(y,x);
t.setFont(f);
t.setBackground(Color.BLACK);
t.setForeground(Color.WHITE);
if(pa.equals("Admin"))
{
    f1 = new Font("Lucida Fax", Font.BOLD, 20);

    l1 = new JLabel("Delete Marks Record ?");
    l1.setForeground(Color.YELLOW);
    l1.setFont(f1);

    l2 = new JLabel("Exam ID");
    l2.setForeground(Color.GRAY);
    l2.setFont(f1);

    ch1 = new Choice();
    ch1.add("Select Exam ID");
    ch1.setFont(f1);

    ConnectionClass obj2 = new ConnectionClass();
    ResultSet rs2 = obj2.stm.executeQuery("SELECT exam_id FROM marks");

    while(rs2.next())
    {
        ch1.add(rs2.getString("exam_id"));
    }

    bt1 = new JButton("Delete Marks");
```

```
bt1.setFont(f1);
bt1.setForeground(Color.RED);
bt1.setBackground(Color.BLACK);

bt1.addActionListener(new ActionListener()
{
    public void actionPerformed(ActionEvent e)
    {
        String exam_id = ch1.getSelectedItem();

        if(exam_id.equals("Select Exam ID"))
        {
            JOptionPane.showMessageDialog(null,"Please select a record");
            return;
        }

        int confirm = JOptionPane.showConfirmDialog(null,
                "Delete record ID: "+exam_id+" ?",
                "Confirm Delete",
                JOptionPane.YES_NO_OPTION);

        if(confirm != JOptionPane.YES_OPTION)
            return;

        try
        {
            ConnectionClass obj3 = new ConnectionClass();
            String q = "DELETE FROM marks WHERE exam_id='"+exam_id+"'";
            int res = obj3.stm.executeUpdate(q);

            if(res==1)
            {
                JOptionPane.showMessageDialog(null,"Marks record deleted");
                setVisible(false);
                new ViewMarksDetails(pu,pa).setVisible(true);
            }
        }
        catch(Exception ex)
        {
            ex.printStackTrace();
        }
    }
});
```

```
            p1 = new JPanel(new GridLayout(1,1));
            p1.add(l1);

            p2 = new JPanel(new GridLayout(1,3,10,10));
            p2.add(l2);
            p2.add(ch1);
            p2.add(bt1);

            p3 = new JPanel(new GridLayout(2,1));
            p3.add(p1);
            p3.add(p2);

            p1.setBackground(Color.BLACK);
            p2.setBackground(Color.BLACK);
            p3.setBackground(Color.BLACK);

            add(p3, BorderLayout.SOUTH);
          }
        }
        catch(Exception ex)
        {
          ex.printStackTrace();
        }

        JScrollPane sp=new JScrollPane(t);
        add (sp);
      }
//    public static void main(String[] args)
//    {
//        new ViewMarksDetails(pu, pa).setVisible(true);
//    }
}
```

AddFeeStructure.java

```
package School_Management_System;

import java.awt.event.*;
import java.awt.*;
import javax.swing.*;
import java.sql.*;
import java.util.Random;

public class AddFeeStructure extends JFrame implements ActionListener
{
    JLabel l1, l2, l3, l4;
    Choice ch1;
    JTextField tf1;
    JButton bt1, bt2;

    AddFeeStructure()
    {
        setTitle("Add Fee Details");
        setSize(800, 560);
        setLocationRelativeTo(null);
        setLayout(null);

        // ---- Background ----
        ImageIcon img = new
ImageIcon(getClass().getResource("/School_Management_System/Icons/feestructure.jpg"));
        Image scaledImg = img.getImage().getScaledInstance(800, 560,
Image.SCALE_SMOOTH);
        JLabel bg = new JLabel(new ImageIcon(scaledImg));
        bg.setBounds(0, 0, 800, 560);
        add(bg);

        // ---- Transparent Panel on top of Background ----
        JPanel panel = new JPanel(null);
        panel.setBounds(200, 100, 400, 300);
        panel.setOpaque(false);
        bg.add(panel);

        // ---- Title ----
        l2 = new JLabel("Add Fee Structure");
        l2.setBounds(80, 0, 300, 40);
        l2.setFont(new Font("Arial", Font.BOLD, 22));
        l2.setForeground(Color.WHITE);
        panel.add(l2);
```

```
// ---- Class Name ----
l3 = new JLabel("Class Name:");
l3.setBounds(30, 70, 150, 30);
l3.setFont(new Font("Arial", Font.BOLD, 18));
l3.setForeground(Color.WHITE);
panel.add(l3);

ch1 = new Choice();
ch1.add("Select Grade");
try {
   ConnectionClass obj = new ConnectionClass();
   String q = "select distinct class_name from class";
   ResultSet rest = obj.stm.executeQuery(q);
   while (rest.next()) {
      ch1.add(rest.getString("class_name"));
   }
} catch (Exception ex) {
   ex.printStackTrace();
}
ch1.setBounds(200, 70, 150, 25);
panel.add(ch1);

// ---- Fee ----
l4 = new JLabel("Fee/Year (Rs):");
l4.setBounds(30, 120, 200, 30);
l4.setFont(new Font("Arial", Font.BOLD, 18));
l4.setForeground(Color.WHITE);
panel.add(l4);

tf1 = new JTextField();
tf1.setBounds(200, 120, 150, 30);
tf1.setFont(new Font("Arial", Font.PLAIN, 18));
panel.add(tf1);

// ---- Buttons ----
bt1 = new JButton("Add Fee");
bt1.setBounds(40, 200, 130, 40);
bt1.setBackground(new Color(0, 0, 0));
bt1.setForeground(Color.WHITE);
bt1.setFont(new Font("Arial", Font.BOLD, 15));
bt1.addActionListener(this);
panel.add(bt1);
```

```
    bt2 = new JButton("Back");
    bt2.setBounds(210, 200, 130, 40);
    bt2.setBackground(new Color(88, 245, 174));
    bt2.setForeground(Color.BLACK);
    bt2.setFont(new Font("Arial", Font.BOLD, 15));
    bt2.addActionListener(this);
    panel.add(bt2);

    setVisible(true);
  }
  @Override
  public void actionPerformed(ActionEvent e)
  {
    if(e.getSource()==bt1)
    {
      String class_name=ch1.getSelectedItem();
      String fee_rs=tf1.getText();
      Random r=new Random();
      String fee_id=""+Math.abs(r.nextInt()%100000);

      if(fee_rs.isEmpty())
      {
        JOptionPane.showMessageDialog(null, "Fee amount should not be an empty");
      }
      else if (Integer.parseInt(fee_rs)>12000)
      {
        JOptionPane.showMessageDialog(null, "Fee amount should not be greater than
12000 rs");
      }
      else if (Integer.parseInt(fee_rs)<10000)
      {
        JOptionPane.showMessageDialog(null, "Fee amount should not be lesser than 10000
rs");
      }
      else
      {
        try
        {
          ConnectionClass obj=new ConnectionClass();
          String q="insert into fee_structure
values('"+fee_id+"','"+class_name+"','"+fee_rs+"')";
          obj.stm.executeUpdate(q);
          JOptionPane.showMessageDialog(null, "Fee Details Successful Inserted");
        }
```

```
                catch(Exception ex)
                {
                    ex.printStackTrace();
                }
            }
        }
        if(e.getSource()==bt2)
        {
            setVisible(false);
        }
    }

//    public static void main(String[] args) {
//        new AddFeeStructure().setVisible(true);
//    }
}
```

AddFeeDetails.java

```
package School_Management_System;

import java.awt.*;
import java.awt.event.*;
import java.sql.*;
import java.util.Random;
import java.util.Date;
import javax.swing.*;

public class AddFeeDetails extends JFrame implements ActionListener
{
   JLabel l1, l2, l3, l4, l5, l6, l7, l8;
   JTextField tf1, tf2, tf3, tf4;
   JButton bt1, bt2;
   Choice ch1, ch2, ch3;

   AddFeeDetails()
   {
      setTitle("Add Student Fee Details");
      getContentPane().setBackground(Color.WHITE);

      setLocation(400,150);
      setSize(900, 600);

      ImageIcon img = new
ImageIcon(getClass().getResource("/School_Management_System/Icons/addfeedetails.jpg"));
      Image scaledImg = img.getImage().getScaledInstance(900, 600,
Image.SCALE_SMOOTH);
      JLabel bg1 = new JLabel(new ImageIcon(scaledImg));
      bg1.setBounds(0, 0, 900, 600);
      add(bg1);

      l1 = new JLabel("Add Student Fee Details", SwingConstants.CENTER);
      l1.setFont(new Font("Arial", Font.BOLD, 32));
      l1.setForeground(new Color(84, 2, 224));
      l1.setBounds(150, 30, 600, 50);
      bg1.add(l1);

      // Class Name
      l2 = new JLabel("Class Name");
      l2.setFont(new Font("Arial", Font.BOLD, 20));
      l2.setForeground(new Color(59, 25, 117));
      l2.setBounds(100, 110, 150, 30);
```

```
bg1.add(l2);

ch1=new Choice();
ch1.add("Select Grade");
try
{
   ConnectionClass obj= new ConnectionClass();
   String q="select distinct class from student";
   ResultSet rest= obj.stm.executeQuery(q);
   while (rest.next())
   {
      ch1.add(rest.getString("class"));
   }
}
catch(Exception ex)
{
   ex.printStackTrace();
}
ch1.setBounds(280, 110, 200, 30);
ch1.setFont(new Font("Arial", Font.BOLD, 20));
bg1.add(ch1);


// Username
l3 = new JLabel("Username");
l3.setFont(new Font("Arial", Font.BOLD, 20));
l3.setBounds(100, 170, 150, 30);
bg1.add(l3);

ch2 = new Choice();
ch2.add("Select Username");
ch2.setBounds(280, 170, 200, 30);
ch2.setFont(new Font("Arial", Font.BOLD, 20));
bg1.add(ch2);

// Student Name
l4 = new JLabel("Student Name");
l4.setFont(new Font("Arial", Font.BOLD, 20));
l4.setBounds(500, 170, 150, 30);
bg1.add(l4);

tf1 = new JTextField();
tf1.setBounds(660, 170, 200, 30);
tf1.setFont(new Font("Arial", Font.BOLD, 20));
```

```java
tf1.setEditable(false);
bg1.add(tf1);

l5 = new JLabel("Email");
l5.setFont(new Font("Arial", Font.BOLD, 20));
l5.setBounds(100, 230, 150, 30);
bg1.add(l5);

tf2 = new JTextField();
tf2.setBounds(280, 230, 200, 30);
tf2.setFont(new Font("Arial", Font.BOLD, 20));
tf2.setEditable(false);
bg1.add(tf2);

// Total Fee
l6 = new JLabel("Total Fee");
l6.setFont(new Font("Arial", Font.BOLD, 20));
l6.setBounds(500, 230, 150, 30);
bg1.add(l6);

tf3 = new JTextField();
tf3.setBounds(660, 230, 200, 30);
tf3.setFont(new Font("Arial", Font.BOLD, 20));
tf3.setEditable(false);
bg1.add(tf3);

// Submit Fee
l7 = new JLabel("Submit Fee");
l7.setFont(new Font("Arial", Font.BOLD, 20));
l7.setBounds(100, 290, 150, 30);
bg1.add(l7);

tf4 = new JTextField();
tf4.setBounds(280, 290, 200, 30);
tf4.setFont(new Font("Arial", Font.BOLD, 20));
bg1.add(tf4);

// Fee Status
l8 = new JLabel("Fee Status");
l8.setFont(new Font("Arial", Font.BOLD, 20));
l8.setBounds(500, 290, 150, 30);
bg1.add(l8);

ch3 = new Choice();
```

```
        ch3.add("Select Status");
        ch3.add("Due");
        ch3.add("Completed");
        ch3.setBounds(660, 290, 200, 30);
        ch3.setFont(new Font("Arial", Font.BOLD, 20));
        bg1.add(ch3);

        bt1 = new JButton("Submit");
        bt1.setBounds(300, 360, 150, 40);
        bt1.setBackground(Color.BLACK);
        bt1.setForeground(Color.WHITE);
        bg1.add(bt1);
        bt1.addActionListener(this);

        bt2 = new JButton("Back");
        bt2.setBounds(500, 360, 150, 40);
        bt2.setBackground(new Color(8, 161, 92));
        bt2.setForeground(Color.BLACK);
        bg1.add(bt2);
        bt2.addActionListener(this);


        ch1.addItemListener(new ItemListener()
        {
          @Override
          public void itemStateChanged(ItemEvent e) {

            if (e.getStateChange() == ItemEvent.SELECTED) {

              String class_name = ch1.getSelectedItem();

              try {

                ch2.removeAll();
                ch2.add("Select Username");

                ConnectionClass objUser = new ConnectionClass();

                String qUser = "SELECT username FROM student WHERE `class`='" +
class_name + "'";
                ResultSet rsUser = objUser.stm.executeQuery(qUser);

                while (rsUser.next()) {
                  ch2.add(rsUser.getString("username"));
```

```
                }

            } catch (Exception ex) {
                ex.printStackTrace();
            }
        }
    }
});
ch2.addItemListener(new ItemListener()
{
    @Override
    public void itemStateChanged(ItemEvent e)
    {
        if (e.getStateChange() == ItemEvent.SELECTED)
        {
            String username = ch2.getSelectedItem();
            String class_name = ch1.getSelectedItem();
            try {
                ConnectionClass obj = new ConnectionClass();
                // Get student name and email
                String q = "SELECT name, email FROM student WHERE username='" +
username + "'";
                ResultSet rs = obj.stm.executeQuery(q);

                if (rs.next()) {
                    tf1.setText(rs.getString("name"));
                    tf2.setText(rs.getString("email"));
                }
                // Get full fee from fee_structure
                String qFee = "SELECT fee_rs FROM fee_structure WHERE class_name='" +
class_name + "'";
                ResultSet rsFee = obj.stm.executeQuery(qFee);

                if (rsFee.next())
                {
                    int fullFee = rsFee.getInt("fee_rs");

                    int completed = 0;
                    int due = 0;

                    // Get completed fees
                    String qCompleted = "SELECT SUM(submited_fee) FROM student_fee
WHERE username='" + username + "' AND status='Completed'";
                    ResultSet rsCompleted = obj.stm.executeQuery(qCompleted);
```

```
                    if(rsCompleted.next())
                        completed = rsCompleted.getInt(1);

                    // Get due fees
                    String qDue = "SELECT SUM(submited_fee) FROM student_fee WHERE
username='" + username + "' AND status='Due'";
                    ResultSet rsDue = obj.stm.executeQuery(qDue);

                    if(rsDue.next())
                        due = rsDue.getInt(1);

                    int remaining = fullFee - completed + due;

                    if(remaining < 0)
                        remaining = 0;

                    tf3.setText(String.valueOf(remaining));
                }

            } catch (Exception ex) {
                ex.printStackTrace();
            }
        }
    }
});

revalidate();
repaint();
setVisible(true);
}

@Override
public void actionPerformed(ActionEvent e)
{
    if(e.getSource()==bt1)
    {
        String class_name=ch1.getSelectedItem();
        String username=ch2.getSelectedItem();
        String name=tf1.getText();
        String email=tf2.getText();
        String total_fee=tf3.getText();
        String submit_fee=tf4.getText();
        String status=ch3.getSelectedItem();
```

```
            Random r=new Random();
            String fee_id=""+Math.abs(r.nextInt() % 100000);
            Date date=new Date();

            if (username == null || username.isEmpty())
            {
               JOptionPane.showMessageDialog(null, "Please select a username!");
               return;
            }
            else if (submit_fee.isEmpty())
            {
               JOptionPane.showMessageDialog(null, "Fee should not be empty!");
            }
            else if (Integer.parseInt(submit_fee) > Integer.parseInt(total_fee))
            {
               JOptionPane.showMessageDialog(null, "Submitted fee cannot be greater than total
fee: " + total_fee);
            }
            else if (Integer.parseInt(submit_fee) == Integer.parseInt(total_fee) &&
                   status.equalsIgnoreCase("Due"))
            {
               JOptionPane.showMessageDialog(null, "If fee is fully paid, status must be Complete
— not Due!");
            }
            else
            {
               try
               {
                  ConnectionClass obj= new ConnectionClass();
                  String q="insert into student_fee values('"+fee_id+"', '"+class_name+"',
'"+username+"', '"+name+"', '"+email+"', '"+total_fee+"', '"+submit_fee+"', '"+status+"',
'"+date+"')";
                  obj.stm.executeUpdate(q);
                  JOptionPane.showMessageDialog(null, "Details Successfully Inserted");
                  this.setVisible(false);
               }
               catch (Exception ee)
               {
                  ee.printStackTrace();
               }
            }
         }
         if (e.getSource()==bt2)
         {
```

```
            setVisible(false);
        }
    }

//    public static void main(String[] args) {
//        new AddFeeDetails().setVisible(true);
//    }
}
```

ViewFeeDetails.java

```
package School_Management_System;

import java.awt.*;
import javax.swing.*;
import java.awt.event.*;
import java.sql.*;

public class ViewFeeDetails extends JFrame
{
   String x[]={"Id", "Class Name", "Username", "Name", "Email", "Total Fee", "Submitted Fee",
"Status", "Date"};
   String y[][]=new String [21][9];
   int i=0,j=0;
   JTable t;
   String query;
   public static String pu, pa;
   Choice ch1;
   JButton bt1;
   JPanel p1, p2, p3;
   JLabel l1, l2;
   Font f, f1;

   ViewFeeDetails(String pub_username, String account2)
   {

      super("Fee Information");
      setSize(1500, 400);
      setResizable(false);
      setLocation(1,1);
      f=new Font("MS UI Gothic", Font.BOLD, 15);

      pu=pub_username;
      pa=account2;
      try
      {
         ConnectionClass obj=new ConnectionClass();
         if(pa.equals("Student"))
         {
            query="select * from student_fee where username='"+pu+"'";
         }
         else if(pa.equals("Admin"))
         {
            query="select * from student_fee";
```

```
}
ResultSet rest=obj.stm.executeQuery(query);
while(rest.next())
{
   y[i][j++]=rest.getString("fee_id");
   y[i][j++]=rest.getString("class_name");
   y[i][j++]=rest.getString("username");
   y[i][j++]=rest.getString("name");
   y[i][j++]=rest.getString("email");
   y[i][j++]=rest.getString("total_fee");
   y[i][j++]=rest.getString("submited_fee");
   y[i][j++]=rest.getString("status");
   y[i][j++]=rest.getString("date");
   i++;
   j=0;
}

t=new JTable(y,x);
t.setFont(f);
t.setBackground(Color.BLACK);
t.setForeground(Color.WHITE);
if(pa.equals("Admin"))
{
   f1 = new Font("Lucida Fax", Font.BOLD, 20);

   l1 = new JLabel("Delete Fee Record ?");
   l1.setForeground(Color.YELLOW);
   l1.setFont(f1);

   l2 = new JLabel("Fee ID");
   l2.setForeground(Color.GRAY);
   l2.setFont(f1);

   ch1 = new Choice();
   ch1.add("Select Fee ID");
   ch1.setFont(f1);

   ConnectionClass obj2 = new ConnectionClass();
   ResultSet rs2 = obj2.stm.executeQuery("SELECT fee_id FROM student_fee");

   while(rs2.next())
   {
      ch1.add(rs2.getString("fee_id"));
   }
```

```java
bt1 = new JButton("Delete Fee");
bt1.setFont(f1);
bt1.setForeground(Color.RED);
bt1.setBackground(Color.BLACK);

bt1.addActionListener(new ActionListener()
{
   public void actionPerformed(ActionEvent e)
   {
      String fee_id = ch1.getSelectedItem();

      if(fee_id.equals("Select Fee ID"))
      {
         JOptionPane.showMessageDialog(null,"Please select a record");
         return;
      }

      int confirm = JOptionPane.showConfirmDialog(null,
            "Delete Fee Record ID: "+fee_id+" ?",
            "Confirm Delete",
            JOptionPane.YES_NO_OPTION);

      if(confirm != JOptionPane.YES_OPTION)
         return;

      try
      {
         ConnectionClass obj3 = new ConnectionClass();
         String q = "DELETE FROM student_fee WHERE fee_id='"+fee_id+"'";
         int res = obj3.stm.executeUpdate(q);

         if(res == 1)
         {
            JOptionPane.showMessageDialog(null,"Fee record deleted!");
            setVisible(false);
            new ViewFeeDetails(pu,pa).setVisible(true);
         }
      }
      catch(Exception ex)
      {
         ex.printStackTrace();
      }
   }
```

```
            });

            p1 = new JPanel(new GridLayout(1,1));
            p1.add(l1);

            p2 = new JPanel(new GridLayout(1,3,10,10));
            p2.add(l2);
            p2.add(ch1);
            p2.add(bt1);

            p3 = new JPanel(new GridLayout(2,1));
            p3.add(p1);
            p3.add(p2);

            p1.setBackground(Color.BLACK);
            p2.setBackground(Color.BLACK);
            p3.setBackground(Color.BLACK);

            add(p3, BorderLayout.SOUTH);
          }
        }
        catch(Exception ex)
        {
          ex.printStackTrace();
        }
        JScrollPane sp = new JScrollPane(t);
        add(sp);
      }
//    public static void main(String[] args)
//    {
//        new ViewFeeDetails(pu, pa).setVisible(true);
//    }
}
```

ShowResult.java

```
package School_Management_System;

import java.awt.*;
import javax.swing.*;
import java.awt.event.*;
import java.sql.*;

public class ShowResult extends JFrame implements ActionListener, ItemListener
{
   JButton bt1, bt2;
   JLabel l1, l2;
   JTextArea ta;
   Choice ch1, ch2;
   JPanel p1;
   Font f;
   String q;
   public static String pu, pa;

   ShowResult(String pub_username, String account2)
   {
      super("Generate Result");

      setSize(500,500);
      setLocation(100,100);
      setResizable(false);

      f = new Font("Arial", Font.BOLD, 16);

      pu = pub_username;
      pa = account2;

      l1 = new JLabel("Class");
      l1.setFont(f);

      ch1 = new Choice();
      ch1.add("Select Grade");

      l2 = new JLabel("Username");
      l2.setFont(f);

      ch2 = new Choice();
      ch2.add("Select Username");
```

```
try
{
    ConnectionClass obj = new ConnectionClass();

    if(pa.equals("Admin") || pa.equals("Teacher"))
    {
        q = "select distinct class_name from marks";
        ResultSet rest = obj.stm.executeQuery(q);

        while(rest.next())
        {
            ch1.add(rest.getString("class_name"));
        }
    }
    else if(pa.equals("Student"))
    {
        q = "select class_name from student where username='"+pu+"'";
        ResultSet rest = obj.stm.executeQuery(q);

        while(rest.next())
        {
            ch1.add(rest.getString("class_name"));
        }
    }

}
catch(Exception ex)
{
    ex.printStackTrace();
}

ch1.setFont(f);
ch2.setFont(f);

// IMPORTANT: add listener to detect class change
ch1.addItemListener(this);

bt1 = new JButton("Show");
bt1.setBackground(Color.BLACK);
bt1.setForeground(Color.WHITE);
bt1.addActionListener(this);

bt2 = new JButton("Print PDF");
bt2.setBackground(Color.BLACK);
```

```
    bt2.setForeground(Color.WHITE);
    bt2.addActionListener(this);

    ta = new JTextArea();
    ta.setFont(f);
    ta.setEditable(false);

    JScrollPane sp = new JScrollPane(ta);

    ta.setText("---------------------------------SCHOOL RESULT---------------------------------");

    p1 = new JPanel();
    p1.setLayout(new GridLayout(4,2,10,10));

    p1.add(l1);
    p1.add(ch1);
    p1.add(l2);
    p1.add(ch2);
    p1.add(bt1);
    p1.add(bt2);

    setLayout(new BorderLayout());
    add(sp, BorderLayout.CENTER);
    add(p1, BorderLayout.SOUTH);
  }

  // THIS METHOD RUNS WHEN CLASS IS SELECTED
  public void itemStateChanged(ItemEvent e)
  {
    if(e.getSource() == ch1)
    {
      ch2.removeAll();
      ch2.add("Select Username");

      String grade = ch1.getSelectedItem();

      try
      {
        ConnectionClass obj = new ConnectionClass();

        if(pa.equals("Admin") || pa.equals("Teacher"))
        {
          q = "select distinct username from marks where class_name='"+grade+"'";
          ResultSet rest = obj.stm.executeQuery(q);
```

```
                while(rest.next())
                {
                    ch2.add(rest.getString("username"));
                }
            }
            else if(pa.equals("Student"))
            {
                q = "select username from marks where username='"+pu+"' and
class_name='"+grade+"'";
                ResultSet rest = obj.stm.executeQuery(q);

                while(rest.next())
                {
                    ch2.add(rest.getString("username"));
                }
            }

        }
        catch(Exception ex)
        {
            ex.printStackTrace();
        }
    }
}

public void actionPerformed(ActionEvent e)
{
    int total = 0;

    if(e.getSource() == bt1)
    {
        ta.setText("----------------------------------SCHOOL RESULT----------------------------------");

        try
        {
            ConnectionClass obj = new ConnectionClass();

            String class_name = ch1.getSelectedItem();
            String username = ch2.getSelectedItem();

            q = "select * from marks where class_name='"+class_name+"' and
username='"+username+"'";
            ResultSet rest1 = obj.stm.executeQuery(q);
```

```
            while(rest1.next())
            {
               ta.append("\n\nStudent Name : " + rest1.getString("name"));
               ta.append("\nTerm : " + rest1.getString("term"));
               ta.append("\n-----------------------------------");

               ta.append("\n" + rest1.getString("subject_name") + " : " + rest1.getString("marks"));

               total = Integer.parseInt(rest1.getString("marks")) + total;
            }

            float percentage = (total * 100f) / 300f;

            ta.append("\n-----------------------------------");
            ta.append("\nTotal Marks : " + total);
            ta.append("\nPercentage : " + percentage + "%");
         }
         catch(Exception ex)
         {
            ex.printStackTrace();
         }
      }

      if(e.getSource() == bt2)
      {
         JOptionPane.showMessageDialog(this, "PDF Printing to be added in the future
upgrade");
      }
   }
}
```

PasswordUtils.java

```
package School_Management_System;

import java.security.*;
import java.util.Base64;
public class PasswordUtils
{
    // Method to hash the password with a salt
    public static String hashPassword(String password) throws NoSuchAlgorithmException
    {
        // Create a salt
        SecureRandom random = new SecureRandom();
        byte[] salt = new byte[16];
        random.nextBytes(salt);
        // Hash the password with the salt
        MessageDigest digest = MessageDigest.getInstance("SHA-256");
        digest.update(salt); // Salt
        byte[] hashedPassword = digest.digest(password.getBytes());
        // Convert the hashed password to a Base64 string for storage
        String hashedPasswordString = Base64.getEncoder().encodeToString(hashedPassword);
        String saltString = Base64.getEncoder().encodeToString(salt);
        // Return a combination of the salt and hashed password to store in the DB
        return saltString + ":" + hashedPasswordString;
    }
    // Method to check if the password matches
    public static boolean verifyPassword(String storedPassword,String originalPassword ) throws
NoSuchAlgorithmException {
    // Check if storedPassword contains a colon
    if (storedPassword == null || !storedPassword.contains(":")) {
        throw new IllegalArgumentException("Stored password format is invalid. It should be in the
format: salt:hashedPassword");
    }
    String[] parts = storedPassword.split(":");
    if (parts.length != 2) {
        throw new IllegalArgumentException("Stored password format is incorrect. It should be in
the format: salt:hashedPassword");
    }
    String saltString = parts[0];
    String hashedPasswordString = parts[1];
    // Convert the salt back from Base64
    byte[] salt = Base64.getDecoder().decode(saltString);
    // Hash the original password with the same salt
    MessageDigest digest = MessageDigest.getInstance("SHA-256");
    digest.update(salt);
```

```
    byte[] hashedPassword = digest.digest(originalPassword.getBytes());
    // Compare the hashed password
    String hashedOriginalPasswordString =
Base64.getEncoder().encodeToString(hashedPassword);
    return hashedOriginalPasswordString.equals(hashedPasswordString);
}
}
```